University College London, 2025-2026
Econ 0039 – Advanced Macro – Franck Portier

Problem set 4: The Liquidity Trap

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Part I – The Krugman Liquidity Trap model with more general preferences

We consider the Krugman model studies in class (endowment economy, cash-in-advance, representative agent). $M_t = M^\star$ for all $t \geq 1$ and $y_t = y^\star$ for all $t \geq 0$. The only difference is that we assume now that preferences are given by:

$$U = \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma} - 1}{1-\sigma}$$

Budget and Cash in Advance constraints are:
Morning: Asset market

$$B_t + \widetilde{M}_t \leq (1 + i_{t-1})B_{t-1} + M_{t-1} + X_t \qquad (\lambda_t)$$

Afternoon: Good market

$$P_t c_t \leq \widetilde{M}_t \qquad (\mu_t)$$
$$M_t \leq \left(\widetilde{M}_t - P_t c_t\right) + P_t \bar{c}_t \qquad (\nu_t)$$

X_t is newly created money distributed to Hh. λ_t, μ_t and ν_t are positive or nul Lagrange multipliers.

We consider first the case in which prices are flexible.

1 – Give the first order conditions of the (representative) household problem, for given prices P and i, not forgetting the complementary slackness conditions.

2 – Prove that λ and ν are always positive.

3 – Why do we assume $i_t \geq 0$?

4 – Prove that μ_t is strictly positive if $i_t > 0$.

5 – What are the three market equilibrium conditions?

6 – Assume that endowments and money supply are constant in period 1 and onwards at level $y^\star$ and $M^\star$, solve for equilibrium for 1 and onwards.

7 – Show that the cash-in-advance constraint might or might not bind in period zero. What is the condition under which it is binding?

8 – Assume that prices are sticky in period 0, such that consumption c can be lower than endowment y. Assume $X_0 = 0$. Under which condition is there a liquidity trap and under-consumption ($c_0 < y^\star$)? Is $X_0 > 0$ a way to increase consumption?

9 – How can the commitment to increase money supply in period 1 can restore efficiency of monetary policy in period 0?

Part II – Chapter 15 of the Keynes' General Theory on the demand for money (document I)

1 – Why is liquidity-preference a better term than income-velocity of money?

2 – What are the four motives for holding money?

3 – In which respect is the speculation motive different from the three other ones?

4 – Summarize the discussion of Keynes on heterogeneity that can be found at the end of section I.

5 – Is Y nominal or real income?

6 – How does money affect the interest rate (top of section II)

7 – Where does Keynes introduces the liquidity trap. What is his line of reasoning?

8 – Why are expectations so important?

9 – What does Keynes mean by *absolute liquidity-preference*?

10 – What are the examples of financial markets breakdowns given by Keynes? Are they about the liquidity trap?

Part III – Krugman and the liquidity trap (documents II to IV)

1 – Krugman writes "creating inflation is easy if you're an irresponsible country. It may not be easy at all if you aren't." What does it mean? Why does it matter?

2 – Krugman writes: " What a model with all the i's dotted and t's crossed actually says is that the CPI doubles if you double the current money supply and all future expected money supplies." Is that true in the model of Part I?

3 – Using the model of Part I, explain why a fiscal fiscal stimulus can be the solution in the liquidity trap.

4 – What does Krugman call *the zero bound*?

5 – What is different in the liquidity trap and outside of it?

Part IV – Some further readings (documents V to IX)

Document I – The General Theory of Employment, Interest and Money, John Maynard Keynes, 1936, Chapter 15: The Psychological and Business Incentives To Liquidity

I

We must now develop in more detail the analysis of the motives to liquidity-preference which were introduced in a preliminary way in Chapter 13. The subject is substantially the same as that which has been sometimes discussed under the heading of the Demand for Money. It is also closely connected with what is called the income-velocity of money; - for the income-velocity of money merely measures what proportion of their incomes the public chooses to hold in cash, so that an increased income-velocity of money may be a symptom of a decreased liquidity-preference. It is not the same thing, however, since it is in respect of "his stock of accumulated savings, rather than of his income, that the individual can exercise his choice between liquidity and illiquidity. And, anyhow, the term "income-velocity of money" carries with it the misleading suggestion of a presumption in favour of the demand for money as' a whole being proportional, or having some determinate relation, to income, whereas this presumption should apply, as we shall see, only to a portion of the public's cash holdings; with the result that it overlooks the part played by the rate of interest.

In my Treatise on Money I studied the total demand for money under the headings of income-deposits, business-deposits, and savings-deposits, and I need not repeat here the analysis which I gave in Chapter 3 of that book. Money held for each of the three purposes forms, nevertheless, a single pool, which the holder is under no necessity to segregate into three water-tight compartments; for they need not be sharply divided even in his own mind, and the same sum can be held primarily for one purpose and secondarily for another. Thus we can - equally well, and, perhaps, better - consider the individual's aggregate demand for money in given circumstances as a single decision, though the composite result of a number of different motives.

In analysing the motives, however, it is still convenient to classify them under certain headings, the first of which broadly corresponds to the former classification of income-deposits and business-deposits, and the two latter to that of savings-deposits. These I have briefly introduced in Chapter 13 under the headings of the transactions-motive, which can be further classified as the income-motive and the business-motive, the precautionary-motive and the speculative-motive.

(i) The Income-motive. - One reason for holding cash is to bridge the interval between the receipt of income and its disbursement. The strength of this motive in inducing a decision to hold a given aggregate of cash will chiefly depend on the amount of income and the normal length of the interval between its receipt and its disbursement. It is in this connection that the concept of the income-velocity of money is strictly appropriate.

(ii) The Business-motive. - Similarly, cash is held to bridge the interval between the time of incurring business costs and that of the receipt of the sale-proceeds; cash held by dealers to bridge the interval between purchase and realisation being included under this heading. The strength of this demand will chiefly depend on the value of current output (and hence on current income), and on the number of hands through which output passes.

(iii) The Precautionary-motive. - To provide for contingencies requiring sudden expenditure and for unforeseen opportunities of advantageous purchases, and also to hold an asset of which the value is fixed in terms of money to meet a subsequent liability fixed in terms of money, are further motives for holding cash.

The strength of all these three types of motive will partly depend on the cheapness and the reliability of methods of obtaining cash, when it is required, by some form of temporary borrowing, in particular by overdraft or its equivalent. For there is no necessity to hold idle cash to bridge over intervals if it can be obtained without difficulty at the moment when it is actually required. Their strength will also depend on what we may term the relative cost of holding cash. If the cash can only be retained by forgoing the purchase of a profitable asset, this increases the cost and thus weakens the motive towards holding a given amount of cash. If deposit interest is earned or if bank charges are avoided by holding cash, this decreases the cost and strengthens the motive. It may be, however, that this is likely to be a minor factor except where large changes in the cost of holding cash are in question.

(iv) There remains the Speculative-motive. - This needs a more detailed examination than the others, both because it is less well understood and because it is particularly important in transmitting the effects of a change in the quantity of money.

In normal circumstances the amount of money required to satisfy the transactions-motive and the precautionary-motive is mainly a resultant of the general activity of the economic system and of the level of money-income. But it is by playing on the speculative-motive that monetary management (or, in the absence of management, chance changes in the quantity of money) is brought to bear on the economic system. For the demand for money to satisfy the former motives is generally irresponsive to any influence except the actual occurrence of a change in the general economic activity and the level of incomes; whereas experience indicates that the aggregate demand for money to satisfy the speculative-motive usually shows a continuous response to gradual changes in the rate of interest, i.e. there is a continuous curve relating changes in the demand for money to satisfy the speculative motive and changes in the rate of interest as given by changes in the prices of bonds and debts of various maturities.

Indeed, if this were not so, "open market operations" would be impracticable. I have said that experience indicates the continuous relationship stated above, because in normal circumstances the banking system is in fact always able to purchase (or sell) bonds in exchange for cash by bidding the price of bonds up (or down) in the market by a modest amount; and the larger the quantity of cash which they seek to create (or cancel) by purchasing (or selling) bonds and debts, the greater must be the fall (or rise) in the rate of interest. Where, however, (as in the United States, 1933-1934) open-market operations have been limited to the purchase of very short-dated securities, the effect may, of course, be mainly confined to the very short-term rate of interest and have but little reaction on the much more important long-term rates of interest.

In dealing with the speculative-motive it is, however, important to distinguish between the changes in the rate of interest which are due to changes in the supply of money available to satisfy the speculative-motive, without there having been any change in the liquidity function, and those which are primarily due to changes in expectation affecting the liquidity function itself. Open-market operations may, indeed, influence the rate of interest through both channels; since they may not only change the volume of money, but may also give rise to changed expectations concerning the future policy of the Central Bank or of the Government. Changes in the liquidity function itself, due to a change in the news which causes revision of expectations, will often be discontinuous, and will, therefore, give rise to a corresponding discontinuity of change in the rate of interest. Only, indeed, in so far as the change in the news is differently interpreted by different individuals or affects individual interests differently will there be room for any increased activity of dealing in the bond market. If the change in the news affects the judgment and the requirements of everyone in precisely the same way, the rate of interest (as indicated by the prices of bonds and debts) will be adjusted forthwith to the new situation without any market transactions being necessary.

Thus, in the simplest case, where everyone is similar and similarly placed, a change in circumstances or expectations will not be capable of causing any displacement of money whatever; - it will simply change the rate of interest in whatever degree is necessary to offset the desire of each individual, felt at the previous rate, to change his holding of cash in response to the new circumstances or expectations; and, since everyone will change his ideas as to the rate which would induce him to alter his holdings of cash in the same degree, no transactions will result. To each set of circumstances and expectations there will correspond an appropriate rate of interest, and there will never be any question of anyone changing his usual holdings of cash.

In general, however, a change in circumstances or expectations will cause some realignment in individual holdings of money; - since, in fact, a change will influence the ideas of different individuals differently by reason partly of differences in environment and the reason for which money is held and partly of differences in knowledge and interpretation of the new situation. Thus the new equilibrium rate of interest will be associated with a redistribution of money-holdings. Nevertheless it is the change in the rate of interest, rather than the redistribution of cash, which deserves our main attention. The latter is incidental to individual differences, whereas the essential phenomenon is that which occurs in the simplest case. Moreover, even in the general case, the shift in the rate of interest is usually the most prominent part of the reaction to a change in the news. The movement in bond-prices is, as the newspapers are accustomed to say, "out of all proportion to the activity of dealing"; - which is as it should be, in view of individuals being much more similar than they are dissimilar in their reaction to news.

II

WHILST the amount of cash which an individual decides to hold to satisfy the transactions-motive and the precautionary-motive is not entirely independent of what he is holding to satisfy the speculative-motive, it is a safe first approximation to regard the amounts of these two sets of cash-holdings as being largely independent of one another. Let us, therefore, for the purposes of our further analysis, break up our problem in this way.

Let the amount of cash held to satisfy the transactions- and precautionary-motives be M_1, and the amount held to satisfy the speculative-motive be M_2. Corresponding to these two compartments of cash, we then have two liquidity functions L_1 and L_2. L_1 mainly depends on the level of income, whilst L_2 mainly depends on the relation between the current rate of interest and the state of expectation. Thus

$$M = M_1 + M_2 = L_1(Y) + L_2(r),$$

where L_1 is the liquidity function corresponding to an income Y, which determines M_1, and L_2 is the liquidity function of the rate of interest r, which determines M_2. It follows that there are three matters to investigate: (i) the relation of changes in M to Y and r, (ii) what determines the shape of L_1, (iii) what determines the shape of L_2.

(i) The relation of changes in M to Y and r depends, in the first instance, on the way in which changes in M come about. Suppose that M consists of gold coins and that changes in M can only result from increased returns to the activities of gold-miners who belong to the economic system under examination. In this case changes in M are, in the first instance, directly associated with changes in Y, since the new gold accrues as someone's income. Exactly the same conditions hold if changes in M are due to the Government printing money wherewith to meet its current expenditure; - in this case also the new money accrues as someone's income. The new level of income, however, will not continue sufficiently high for the requirements of M, to absorb the whole of the increase in M; and some portion of the money will seek an outlet in buying securities or other assets until r has fallen so as to bring about an increase in the magnitude of M, and at the same time to stimulate a rise in Y to such an extent that the new money is absorbed either in M_2 or in the M_1 which corresponds to the rise in Y caused by the fall in r. Thus at one remove this case comes to the same thing as the alternative case, where the new money can only be issued in the first instance by a relaxation of the conditions of credit by the banking system, so as to induce someone to sell the banks a debt or a bond in exchange for the new cash.

It will, therefore, be safe for us to take the latter case as typical. A change in M can be assumed to operate by changing r, and a change in r will lead to a new equilibrium partly by changing M_2 and partly by changing Y and therefore M_1. The division of the increment of cash between M_1 and M_2 in the new position of equilibrium will depend on the responses of investment to a reduction in the rate of interest and of income to an increase in investment.[1] Since Y partly depends on r, it follows that a given change in M has to cause a sufficient change in r for the resultant changes in M_1 and M_2 respectively to add up to the given change in M.

(ii) It is not always made clear whether the income-velocity of money is defined as the ratio of Y to M or as the ratio of Y to M_1. I propose, however, to take it in the latter sense. Thus if V is the income-velocity of money,

$$L_1(Y) = Y/V = M_1.$$

There is, of course, no reason for supposing that V is constant. Its value will depend on the character of banking and industrial organisation, on social habits, on the distribution of income between different classes and on the effective cost of holding idle cash. Nevertheless, if we have a short period of time in view and can safely assume no material change in any of these factors, we can treat V as nearly enough constant.

(iii) Finally there is the question of the relation between M_2 and r. We have seen in Chapter 13 that uncertainty as to the future course of the rate of interest is the sole intelligible explanation of the type of liquidity-preference L_2 which leads to the holding of cash M_2. It follows that a given M_2 will not have a definite quantitative relation to a given rate of interest of r; - what matters is not the absolute level of r but the degree of its divergence from what is considered a fairly safe level of r, having regard to those calculations of probability which are being relied on. Nevertheless, there are two reasons for expecting that, in any given state of expectation, a fall in r will be associated with an increase in M_2. In the first place, if the general view as to what is a safe level of r is unchanged, every fall in r reduces the market rate relatively to the "safe" rate and therefore increases the risk of illiquidity; and, in the second place, every fall in r reduces the current earnings from illiquidity, which are available as a sort of insurance premium to offset the risk of loss on capital account, by an amount equal to the difference between the squares of the old rate of interest and the new. For example, if the rate of interest on a long-term debt is 4 per cent., it is preferable to sacrifice liquidity unless on a balance of probabilities it is feared that the long-term rate of interest may rise faster than by 4 per cent. of itself per annum, i.e. by an amount greater than 0.16 per cent. per annum. If, however, the rate of interest is already as low as 2 per cent., the running yield will only offset a rise in it of as little as 0.04 per cent. per annum. This, indeed, is perhaps the chief obstacle to a fall in the rate of interest to a very low level. Unless reasons are believed to exist why future experience will be very different from past experience, a long-term rate of interest of (say) 2 per cent. leaves more to fear than to hope, and offers, at the same time, a running yield which is only sufficient to offset a very small measure of fear.

It is evident, then, that the rate of interest is a highly psychological phenomenon. We shall find, in equilibrium at a level below the rate which corresponds to full employment; because at such a level a state of true inflation will be produced, with the result that M_1 will absorb ever-increasing quantities of cash. But at a level above the rate

[1] We must postpone to Book V. the question of what will determine the character of the new equilibrium.

which corresponds to full employment, the long-term market-rate of interest will depend, not only on the current policy of the monetary authority, but also on market expectations concerning its future policy. The short-term rate of interest is easily controlled by the monetary authority, both because it is not difficult to produce a conviction that its policy will not greatly change in the very near future, and also because the possible loss is small compared with the running yield (unless it is approaching vanishing point). The the long-term rate may be more recalcitrant when once it has fallen to a level which, on the basis of past experience and present expectations of future monetary policy, is considered "unsafe" by representative opinion. For example, in a country linked to an international gold standard, a rate of interest lower than prevails elsewhere will be viewed with a justifiable lack of confidence; yet a domestic rate of interest dragged up to a parity with the highest rate (highest after allowing for risk) prevailing in any country belonging to the international system may be much higher than is consistent with domestic full employment.

Thus a monetary policy which strikes public opinion as being experimental in character or easily liable to change may fail in its objective of greatly reducing the long-term rate of interest, because M_2 may tend to increase almost without limit in response to a reduction of r below a certain figure. The same policy, on the other hand, may prove easily successful if it appeals to public opinion as being reasonable and practicable and in the public interest, rooted in strong conviction, and promoted by an authority unlikely to be superseded.

It might be more accurate, perhaps, to say that the rate of interest is a highly conventional, rather than a highly psychological, phenomenon. For its actual value is largely governed by the prevailing view as to what its value is expected to be. Any level of interest which is accepted with sufficient conviction as likely to be durable will be durable; subject, of course, in a changing society to fluctuations for all kinds of reasons round the expected normal. In particular, when M_1 is increasing faster than M, the rate of interest will rise, and vice versa. But it may fluctuate for decades about a level which is chronically too high for full employment; - particularly if it is the prevailing opinion that the rate of interest is self-adjusting, so that the level established by convention is thought to be rooted in objective grounds much stronger than convention, the failure of Employment to attain an optimum level being in no way associated, in the minds either of the public or of authority, with the prevalence of an inappropriate range of rates of interest.

The difficulties in the way of maintaining effective demand at a level high enough to provide full employment, which ensue from the association of a conventional and fairly stable long-term rate of interest with a fickle and highly unstable marginal efficiency of capital, should be, by now, obvious to the reader.

Such comfort as we can fairly take from more encouraging reflections must be drawn from the hope that, precisely because the convention is not rooted in secure knowledge, it will not be always unduly resistant to a modest measure of persistence and consistency of purpose by the monetary authority. Public opinion can be fairly rapidly accustomed to a modest fall in the rate of interest and the conventional expectation of the future may be modified accordingly; thus preparing the way for a further movement - up to a point. The fall in the long-term rate of interest in Great Britain after her departure from the gold standard provides an interesting example of this; - the major movements were effected by a series of discontinuous jumps, as the liquidity function of the public, having become accustomed to each successive reduction, became ready to respond to some new incentive in the news or in the policy of the authorities.

III

We can sum up the above in the proposition that in any given state of expectation there is in the minds of the public a certain potentiality towards holding cash beyond what is required by the transactions-motive or the precautionary-motive, which will realise itself in actual cash-holdings in a degree which depends on the terms on which the monetary authority is willing to create cash. It is this potentiality which is summed up in the liquidity function L_2.

Corresponding to the quantity of money created by the monetary authority, there will, therefore, be cet. par. a determinate rate of interest or, more strictly, a determinate complex of rates of interest for debts of different maturities. The same thing, however, would be true of any other factor in the economic system taken separately. Thus this particular analysis will only be useful and significant in so far as there is some specially direct or purposive connection between changes in the quantity of money and changes in the rate of interest. Our reason for supposing that there is such a special connection arises from the fact that, broadly speaking, the banking system and the monetary authority are dealers in money and debts and not in assets or consumables.

If the monetary authority were prepared to deal both ways on specified terms in debts of all maturities, and even more so if it were prepared to deal in debts of varying degrees of risk, the relationship between the complex of rates of interest and the quantity of money would be direct. The complex of rates of interest would simply be an expression of the terms on which the banking system is prepared to acquire or part with debts; and the quantity of money would be the amount which can find a home in the possession of individuals who - after taking account of all relevant circumstances - prefer the control of liquid cash to parting with it in exchange for a debt on the terms indicated by the market rate of interest. Perhaps a complex offer by the central bank to buy and sell at stated prices gilt-edged bonds of all maturities, in place of t"he single bank rate for short-term bills, is the most important practical improvement which can be made in the technique of monetary management.

To-day, however, in actual practice, the extent to which the price of debts as fixed by the banking system is "effective" in the market, in the sense that it governs the actual market-price, varies in different systems. Sometimes the price is more effective in one direction than in the other; that is to say, the banking system may undertake to purchase debts at a certain price but not necessarily to sell them at a figure near enough to its buying-price to represent no more than a dealer's turn, though there is no reason why the price should not be made effective both ways with the aid of open-market operations. There is also the more important qualification which arises out of the monetary authority not being, as a rule, an equally willing dealer in debts of all maturities. The monetary authority often tends in practice to concentrate upon short-term debts and to leave the price of long-term debts to be influenced by belated and imperfect reactions from the price of short-term debts; - though here again there is no reason why they need do so. Where these qualifications operate, the directness of the relation between the rate of interest and the quantity of money is correspondingly modified. In Great Britain the field of deliberate control appears to be widening. But in applying this theory in any particular case allowance must be made for the special characteristics of the method actually employed by the monetary authority. If the monetary authority deals only in short-term debts, we have to consider what influence the price, actual and prospective, of short-term debts exercises on debts of longer maturity.

Thus there are certain limitations on the ability of the monetary authority to establish any given complex of rates of interest for debts of different terms and risks, which can be summed up as follows:

(1) There are those limitations which arise out of the monetary authority's own practices in limiting its willingness to deal to debts of a particular type.

(2) There is the possibility, for the reasons discussed above, that, after the rate of interest has fallen to a certain level, liquidity-preference may become virtually absolute in the sense that almost everyone prefers cash to holding a debt which yields so low a rate of interest. In this event the monetary authority would have lost effective control over the rate of interest. But whilst this limiting case might become practically important in future, I know of no example of it hitherto. Indeed, owing to the unwillingness of most monetary authorities to deal boldly in debts of long term, there has not been much opportunity for a test. Moreover, if such a situation were to arise, it would mean that the public authority itself could borrow through the banking system on an unlimited scale at a nominal rate of interest.

(3) The most striking examples of a complete breakdown of stability in the rate of interest, due to the liquidity function flattening out in one direction or the other, have occurred in very abnormal circumstances. In Russia and Central Europe after the war a currency crisis or flight from the currency was experienced, when no one could be induced to retain holdings either of money or of debts on any terms whatever, and even a high and rising rate of interest was unable to keep pace with the marginal efficiency of capital (especially of stocks of liquid goods) under the influence of the expectation of an ever greater fall in the value of money; whilst in the United States at certain dates in 1932 there was a crisis of the opposite kind - a financial crisis or crisis of liquidation, when scarcely anyone could be induced to part with holdings of money on any reasonable terms.

(4) There is, finally, the difficulty discussed in section iv of Chapter 11, p. 144, in the way of bringing the effective rate of interest below a certain figure, which may prove important in an era of low interest-rates; namely the intermediate costs of bringing the borrower and the ultimate lender together, and the allowance for risk, especially for moral risk, which the lender requires over and above the pure rate of interest. As the pure rate of interest declines it does not follow that the allowances for expense and risk decline pari passu. Thus the rate of interest which the typical borrower has to pay may decline more slowly than the pure rate of interest, and may be incapable of being brought, by the methods of the existing banking and financial organisation, below a certain minimum figure. This is particularly important if the estimation of moral risk is appreciable. For where the risk is due to doubt in the mind of the lender concerning the honesty of the borrower, there is nothing in the mind of a borrower who does not intend to be dishonest to offset the resultant higher charge. It is also important in the case of short-term loans (e.g. bank loans) where the expenses are heavy; - a bank may have to charge its customers 1 1/2 to 2 per cent., even if the pure rate of interest to the lender is nil.

IV

At the cost of anticipating what is more properly the subject of Chapter 21 below it may be interesting briefly at this stage to indicate the relationship of the above to the Quantity Theory of Money.

In a static society or in a society in which for any other reason no one feels any uncertainty about the future rates of interest, the Liquidity Function L_2 or the propensity to hoard (as we might term it), will always be zero in equilibrium. Hence in equilibrium $M_2 = 0$ and $M = M_1$; so that any change in M will cause the rate of interest to fluctuate until income reaches a level at which the change in M_1 is equal to the supposed change in M. Now $M_1 \times V = Y$, where V is the income-velocity of money as defined above and Y is the aggregate income. Thus if it is practicable to measure the quantity, O, and the price, P, of current output, we have $Y = O \times P$, and, therefore, $M \times Y = O \times P$; which is much the same as the Quantity Theory of Money in its traditional form.[2]

[2] If we had defined V, not as equal to Y/M, but as equal to Y/M_1, then, of course, the Quantity Theory is a truism which holds in all circumstances, though without significance.

For the purposes of the real world it is a great fault in the Quantity Theory that it does not distinguish between changes in prices which are a function of changes in output, and those which are a function of changes in the wage-unit.[3] The explanation of this omission is, perhaps, to be found in the assumptions that there is no propensity to hoard and that there is always full employment. For in this case, O being constant and $M2$ being zero, it follows, if we can take V also as constant, that both the wage-unit and the price-level will be directly proportional to the quantity of money.

Document II – Macro policy in a liquidity trap, Paul Krugman - New York Times Blog, November 15, 2008

"Macro policy in a liquidity trap. That's the title of a new report from Jan Hatzius et al at Goldman Sachs (not available online). The Goldman guys, like me, come up with scary figures about the size of the gap in demand that needs to be filled - figures that suggest the need for a fiscal stimulus that's enormous by historical standards. Their approach is different, and probably better than mine; I'll get to that in a bit. But I want to talk conceptual stuff for a moment.

It's a curious thing that even now, when we are clearly in a liquidity trap, we still have a lot of economists denying that such a thing is possible. The argument seems to go like this: creating inflation is easy - birds do it, bees do it, Zimbabwe does it. So it can't really be a problem for competent countries like Japan or the United States.

This misses a key point that I and others tried to make for Japan in the 90s and are trying to make again now: creating inflation is easy *if you're an irresponsible country.* It may not be easy at all if you aren't.

A decade ago, when I tried to make sense of Japan's predicament, I used a simple, unrealistic model to ask what we really know about the relationship between the money supply and the price level. We normally say that an increase in the money supply, other things equal, leads to an equal proportional increase in the price level: double M and you double the CPI. But that's not actually right. What a model with all the i's dotted and t's crossed actually says is that the CPI doubles if you double the current money supply and *all future expected money supplies.*

And how do you do that? No matter how much Japan increases the monetary base now, expectations of future money supplies won't move if people believe that the Bank of Japan will move to stabilize the price level as soon as the economy recovers. And once you realize that central banks may not be able to move expectations about future money supplies, it becomes a real possibility that the economy will be in a liquidity trap: if interest rates are near zero, money printed now just gets hoarded, and monetary policy has no traction on the real economy.

Zimbabwe wouldn't have this problem: people believe that any money it prints will stay in circulation. But the likes of Japan, or the United States, print money for policy purposes, not to pay their bills. And that, perversely, is what makes them vulnerable to a liquidity trap. Back in 1998 I argued that the Bank of Japan needed to find a way to "credibly promise to be irresponsible." That didn't go down too well, but it was what sober, careful economic analysis prescribed.

Or as I said in the linked paper,

> "The whole subject of the liquidity trap has a sort of Alice-through-the-looking-glass quality. Virtues like saving, or a central bank known to be strongly committed to price stability, become vices; to get out of the trap a country must loosen its belt, persuade its citizens to forget about the future, and convince the private sector that the government and central bank aren't as serious and austere as they seem."

OK, so now back to Hatzius et al. They emphasize the role of the disruption of credit markets in pushing us into a liquidity trap. They then turn to an estimate of likely changes in the "private sector balance" - the difference between private sector saving and private sector investment. And it's stunning:

> "The GS house price forecast combined with current equity prices and credit spreads implies a rise in the private sector balance from +1% of GDP in the second quarter of 2008 to +10% in the fourth quarter of 2009 - a rise of 9 percentage points, or 6 points at an annual rate."

What's the answer? Huge fiscal stimulus, to fill the hole. More aggressive GSE lending. Maybe a "pre-commitment" by the Fed to keep rates low for an extended period - that's a more genteel version of my "credibly promise to be irresponsible." And maybe large-scale purchases of risky assets.

The main thing to realize is that for the time being we really are in an alternative universe, in which nothing would be more dangerous than an attempt by policy makers to play it safe.

[3] This point will be further developed in Chapter 21 below.

Document III – Nobody Understands The Liquidity Trap, Paul Krugman - New York Times Blog, July 14, 2010

Sigh. In an otherwise useful article about divisions in the Fed, Jon Hilsenrath says this:

> "The Fed is better equipped to solve some economic problems than others. As Mr. Bernanke noted in a now-famous 2002 speech, the Fed has the power to fight deflation-or falling wages and prices-by printing money.
>
> But the bank's tools aren't perfectly suited to reducing unemployment, which is influenced by a range of factors including fiscal policy, regulation and global demand."

Sorry, but that's totally wrong. The question is whether, at the zero bound, the Fed has the ability to increase aggregate demand - full stop. If it can increase aggregate demand, it can fight both deflation and unemployment; if not, not.

In a way, the problem with Bernanke's speech was that he made increasing demand and fighting deflation sound too easy. The Fed can print money, if you increase the supply of something its price will fall, end of story.

But as I tried to point out a long time ago, this simple story breaks down when short-term interest rates are near zero.

Here's one way to think about it: when the Fed conducts an open-market operation, buying short-term debt with newly printed money, this normally affects the short rate because bonds and money are imperfect substitutes: money yields less, but has the advantage of being something you can use directly to make payments, that is, it's more liquid.

But when you have bought so much debt and created so much money that rates are near zero, the public is saturated with liquidity; from that point on, they're holding money simply as a store of value, which makes it no different from bonds - and hence a perfect substitute for bonds. And at that point further open-market operations do nothing - they just swap one zero-interest asset for another, with no effect on anything.

So why not forget about open-market operations, and just drop the stuff from helicopters? Well, remember that at this point cash and short-term bonds are equivalent. So a helicopter drop is just like a temporary lump-sum tax cut. And we would expect people to save much or most of such a tax cut - all of it, if you believe in full Ricardian equivalence.

In my simple 1998 model, there's only one way the Fed can affect things at all: by promising, credibly, to print more money in the future, when the zero lower bound no longer binds.

In practice, things are more complicated, because long-term bonds aren't perfect substitutes for short-term - so the Fed can get some traction by buying at longer maturities. But I always felt than Ben was overstating the effectiveness of such purchases. It's worth noting that in his "it" speech Bernanke's more-or-less specific proposal was to set a ceiling on the yield on two-year securities. How much would that accomplish now, when even the 2-year yield is only 0.67 percent?

Anyway, back to the original point: it's depressing to realize that two years into liquidity trap economics, the WSJ still doesn't seem to understand the basic point of why the zero bound is a problem.

Document IV – Wage-Price Flexibility in a Liquidity Trap, Again Again Again, Paul Krugman - New York Times Blog, July 15, 2013

One of the frustrating things about macroeconomic discussion since the Great Recession struck is the prevalence of zombie fallacies - misconceptions that one imagines have been killed by logic or evidence, but just keep coming back to eat our brains. Often, maybe usually, politics is what's keeping these zombies alive; or, if not exactly politics, the attempt of economists to defend their intellectual investments in failed theories.

Sometimes, however, the zombies manage to eat a brain or two simply because someone wasn't paying attention. And I think this is what has just happened to the usually excellent Noah Smith.

Smith finds Japan's persistent shortfall puzzling, because - he claims - this isn't supposed to happen in New Keynesian models:

> "In a New Keynesian model, when there is a demand shortfall, unemployment is the result. The central bank can print money in order to combat the shortfall, which raises inflation and lowers unemployment. But if the central bank does nothing, prices will eventually adjust, and unemployment will go away. This New Keynesian model corresponds nicely to the simple AD-AS model that people learn in Econ 102."

In the words of Charlie Brown, aaugh!

People, we've been through this.

Yes, in a standard AS-AD or NK model, high unemployment leads to falling wages and prices, and this eventually restores full employment. But how does this happen? Not because making labor cheaper increases the quantity of labor demanded - Keynes understood that point perfectly long before he even wrote the General Theory:

> "Or again, if a particular producer or a particular country cuts wages, then, so long as others do not follow suit, that producer or that country is able to get more of what trade is going. But if wages are cut all round, the purchasing power of the community as a whole is reduced by the same amount as the reduction of costs; and, again, no one is further forward."

No, the only reason deflation "works" in the standard model is that it increases the real money supply, which leads to lower interest rates; in effect, it acts like an expansionary monetary policy.

But *Japan has been in a liquidity trap* during the whole period Smith looks at.

Monetary expansion is ineffective unless it can raise expectations of future inflation. Deflation is definitely not going to help. In fact, by raising the real burden of debt, it makes things worse A corollary is that while sticky wages are a real phenomenon - the evidence just keeps getting stronger - their importance has to be appreciated correctly. You need them to understand what we're seeing, which is the failure of deflation to appear in the US now (and the slow pace of deflation in Japan). They are not, repeat NOT the reason either Japan or we have failed to recover.

How is it that this stuff - which is more or less where we came in - hasn't gotten through?

(*Further reading*) Document V – Goods and Financial Markets: The IS-LM Model, Chapter 5 in *Macroeconomics, 8th Edition*, Olivier Blanchard, 2021

Goods and Financial Markets: The IS-LM Model

5

In Chapter 3, we looked at the goods market. In Chapter 4, we looked at financial markets. We now look at goods and financial markets together. By the end of this chapter you will have a framework to think about how output and the interest rate are determined in the short run.

In developing this framework, we follow a path first traced by two economists, John Hicks and Alvin Hansen in the late 1930s and early 1940s. When the economist John Maynard Keynes published his *General Theory* in 1936, there was much agreement that his book was both fundamental and nearly impenetrable. (Try to read it, and you will agree.) There were (and still are) many debates about what Keynes "really meant." In 1937, John Hicks summarized what he saw as one of Keynes's main contributions: the joint description of goods and financial markets. His analysis was later extended by Alvin Hansen. Hicks and Hansen called their formalization the *IS-LM* (investment-saving, liquidity-money) model.

Macroeconomics has made substantial progress since the early 1940s. This is why the IS-LM model is treated in this and the next chapter rather than in Chapter 24 of this book. (If you had taken this course 40 years ago, you would be nearly done!) But to most economists, the IS-LM model still represents an essential building block—one that, despite its simplicity, captures much of what happens in the economy in *the short run.* This is why the IS-LM model is still taught and used today.

The version of the IS-LM model presented in this book is a bit different (and, you will be happy to know, simpler) than the model developed by Hicks and Hansen. This reflects a change in the way central banks now conduct monetary policy, with a shift in focus from controlling the money stock to controlling the interest rate, as explained in Chapter 4.

This chapter develops the basic version of the IS-LM model.

Section 5-1 looks at equilibrium in the goods market and derives the IS relation.

Section 5-2 looks at equilibrium in financial markets and derives the LM relation.

Sections 5-3 and 5-4 put the IS and LM relations together and use the resulting IS-LM model to study the effects of fiscal and monetary policy—first separately, then together.

Section 5-5 introduces dynamics and explores how the IS-LM model captures what happens in the economy in the short run.

If you remember one basic message from this chapter, it should be: In the short run, output is determined by equilibrium in the goods and financial markets. ▶▶▶

5-1 THE GOODS MARKET AND THE IS RELATION

Let's first summarize what we learned in Chapter 3:

- We characterized equilibrium in the goods market as the condition that production, Y, be equal to the demand for goods, Z. We called this condition the IS relation.
- We defined demand as the sum of consumption, investment, and government spending. We assumed that consumption was a function of disposable income (income minus taxes), and took investment spending, government spending, and taxes as given:

$$Z = C(Y - T) + \bar{I} + G$$

(In Chapter 3, we assumed, to simplify the algebra, that the relation between consumption, C, and disposable income, $Y-T$, was linear. Here, we shall not make this assumption but use the more general form $C = C(Y-T)$ instead.)

- The equilibrium condition was thus given by

$$Y = C(Y - T) + \bar{I} + G$$

Using this equilibrium condition, we then looked at the factors that moved equilibrium output. We looked in particular at the effects of changes in government spending and of shifts in consumption demand.

Much more on the effects of interest rates on both consumption and investment in Chapter 15.

The main simplification of this first model was that the interest rate did not affect the demand for goods. Our first task in this chapter is to abandon this simplification and introduce the interest rate in our model of equilibrium in the goods market. For the time being, we focus only on the effect of the interest rate on investment and leave the discussion of its effects on the other components of demand until later.

Investment, Sales, and the Interest Rate

In Chapter 3, investment was assumed to be constant. This was for simplicity. Investment is in fact far from constant and depends primarily on two factors:

- The level of sales. Consider a firm facing an increase in sales and needing to increase production. To do so, it may need to buy additional machines or build an additional plant. In other words, it needs to invest. A firm facing low sales will feel no such need and will spend little, if anything, on investment.

The argument still holds if the firm uses its own funds: The higher the interest rate, the more attractive it is to lend the funds rather than to use them to buy the new machine.

- The interest rate. Consider a firm deciding whether to buy a new machine. Suppose that to buy the new machine, the firm must borrow. The higher the interest rate, the less attractive it is to borrow and buy the machine. (For the moment, and to keep things simple, we make two simplifications. First, we assume that all firms can borrow at the same interest rate—namely, the interest rate on bonds as determined in Chapter 4. In fact, many firms borrow from banks, possibly at a different rate. We also leave aside the distinction between the nominal interest rate—the interest rate in terms of dollars—and the real interest rate—the interest rate in terms of goods. We return to both issues in Chapter 6.) At a high enough interest rate, the additional profits from using the new machine will not cover interest payments, and the new machine will not be worth buying.

To capture these two effects, we write the investment relation as follows:

$$I = \underset{(+,-)}{I(Y, i)} \tag{5.1}$$

Equation (5.1) states that investment I depends on production Y and the interest rate i. (We continue to assume that inventory investment is equal to zero, so sales and production are always equal. As a result, Y denotes both sales and production.) The positive sign under Y indicates that an increase in production (equivalently, an increase in sales) leads to an increase in investment. The negative sign under the interest rate i indicates that an increase in the interest rate leads to a decrease in investment.

An increase in output leads to an increase in investment. An increase in the interest rate leads to a decrease in investment.

Determining Output

Taking into account the investment relation (5.1), the condition for equilibrium in the goods market becomes

$$Y = C(Y - T) + I(Y, i) + G \tag{5.2}$$

Production (the left side of the equation) must be equal to the demand for goods (the right side). Equation (5.2) is our expanded IS relation. We can now look at what happens to output when the interest rate changes.

Start with Figure 5-1. Measure the demand for goods (Z) on the vertical axis. Measure output (Y) on the horizontal axis. For a given value of the interest rate i, demand is an increasing function of output, for two reasons:

- An increase in output leads to an increase in income and thus to an increase in disposable income. The increase in disposable income leads to an increase in consumption. We studied this relation in Chapter 3.
- An increase in output also leads to an increase in investment. This is the relation between investment and sales that we have introduced in this chapter.

In short, an increase in output leads, through its effects on both consumption and investment, to an increase in the demand for goods. This relation between demand and output, for a given interest rate, is represented by the upward-sloping curve ZZ.

Note two characteristics of ZZ in Figure 5-1:

- Because we have not assumed that the consumption and investment relations in equation (5.2) are linear, ZZ is in general a curve rather than a line, as shown in Figure 5-1. All the arguments that follow would apply if we assumed

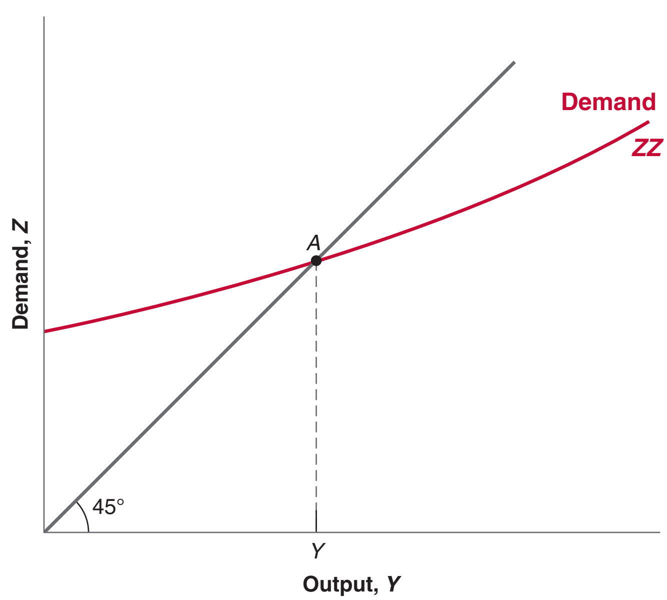


Figure 5-1

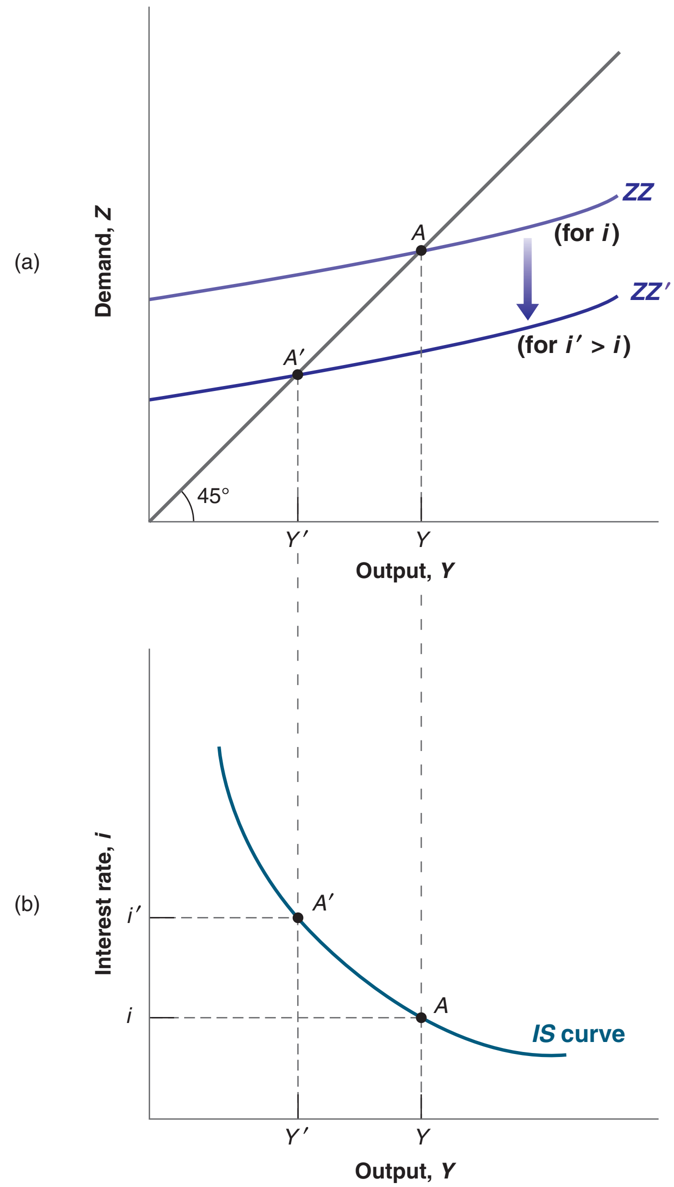
Equilibrium in the Goods Market

The demand for goods is an increasing function of output. Equilibrium requires that the demand for goods be equal to output.

Figure 5-2

The IS Curve

(a) An increase in the interest rate decreases the demand for goods at any level of output, leading to a decrease in the equilibrium level of output. (b) Equilibrium in the goods market implies that an increase in the interest rate leads to a decrease in output. The IS curve, which gives the relation between the interest rate and output, is therefore downward sloping.



that the consumption and investment relations were linear and that ZZ were a straight line.

- We have drawn ZZ so that it is flatter than the 45-degree line. Put another, but equivalent, way, we have assumed that an increase in output leads to a less than one-for-one increase in demand. In Chapter 3, where investment was constant, this restriction naturally followed from the assumption that consumers spend only part of their additional income on consumption. But now that we allow investment to respond to production, this restriction may no longer hold. In response to an increase in output, the sum of the increase in consumption and the increase in investment could exceed the initial increase in output. Although this is a theoretical possibility, the empirical evidence suggests that it is not the case in reality. That's why we shall assume that the response of demand to output is less than one for one and draw ZZ flatter than the 45-degree line.

Make sure you understand ▶ why the two statements mean the same thing.

Equilibrium in the goods market is reached at the point where the demand for goods equals output; that is, at point A, the intersection of ZZ and the 45-degree line. The equilibrium level of output is given by Y.

So far, what we have done is extend, in straightforward fashion, the analysis of Chapter 3. We are now ready to derive the IS curve.

Deriving the IS Curve

We have drawn the demand relation, ZZ, in Figure 5-1 for a given value of the interest rate. Let's now derive in Figure 5-2 what happens if the interest rate changes.

Suppose that, in Figure 5-2(a), the demand curve is given by ZZ and the initial equilibrium is at point A. Suppose now that the interest rate increases from its initial value i to i'. At any level of output, the higher interest rate leads to lower investment and lower demand. The demand curve ZZ shifts down to ZZ': At a given level of output, demand is lower. The new equilibrium is at the intersection of the lower demand curve ZZ' and the 45-degree line, at point A'. The equilibrium level of output is now equal to Y'.

In words: The increase in the interest rate decreases investment. The decrease in investment leads to a decrease in output, which further decreases consumption and investment, through the multiplier effect.

Can you show graphically the size of the multiplier? (*Hint*: Look at the ratio of the decrease in equilibrium output to the initial decrease in investment.)

Using Figure 5-2(a), we can find the equilibrium value of output associated with *any* value of the interest rate. The resulting relation between equilibrium output and the interest rate is drawn in Figure 5-2(b).

Figure 5-2(b) plots equilibrium output Y on the horizontal axis against the interest rate on the vertical axis. Point A in Figure 5-2(b) corresponds to point A in Figure 5-2(a), and point A' in Figure 5-2(b) corresponds to A' in Figure 5-2(a). The higher interest rate is associated with a lower level of output.

This relation between the interest rate and output is represented by the downward-sloping curve in Figure 5-2(b). This curve is called the **IS curve**.

Equilibrium in the goods market implies that an increase in the interest rate leads to a decrease in output. This relation is represented by the downward-sloping IS curve.

Shifts of the IS Curve

We have drawn the IS curve in Figure 5-2, taking as given the values of taxes, T, and government spending, G. Changes in either T or G will shift the IS curve.

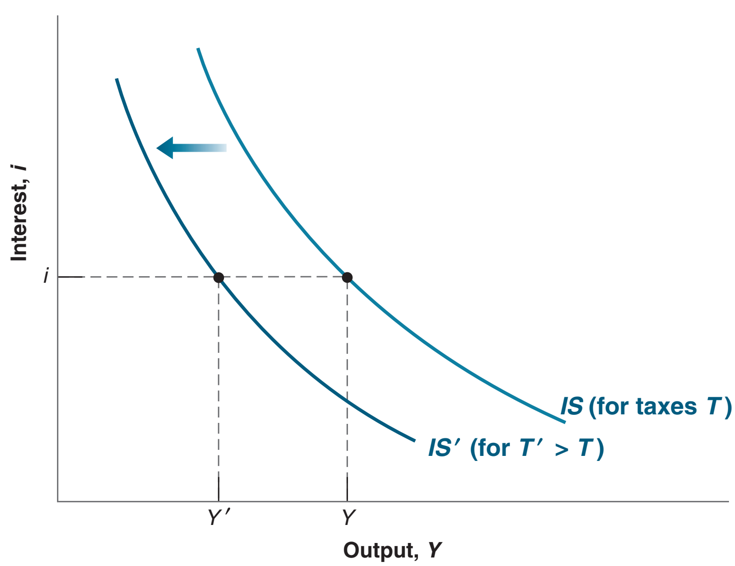


Figure 5-3

Shifts of the IS Curve

An increase in taxes shifts the IS curve to the left.

For a given interest rate, an increase in taxes leads to a decrease in output. In graphic terms: An increase in taxes shifts the IS curve to the left. ▶

To see how, consider Figure 5-3. The IS curve gives the equilibrium level of output as a function of the interest rate. It is drawn for given values of taxes and spending. Now consider an increase in taxes, from T to T'. At a given interest rate, i, disposable income decreases, leading to a decrease in consumption, leading in turn to a decrease in the demand for goods and a decrease in equilibrium output, from Y to Y'. The IS curve shifts to the left: At a given interest rate, the equilibrium level of output is lower than it was before the increase in taxes.

More generally, any factor that, for a given interest rate, decreases the equilibrium level of output causes the IS curve to shift to the left. We have looked at an increase in taxes. But the same would hold for a decrease in government spending or in consumer confidence (which decreases consumption given disposable income). Symmetrically, any factor that, for a given interest rate, increases the equilibrium level of output—a decrease in taxes, an increase in government spending, an increase in consumer confidence—causes the IS curve to shift to the right.

Suppose that the government announces that ▶ the Social Security system is in trouble and it may have to cut retirement benefits in the future. How are consumers likely to react? What is then likely to happen to demand and output?

Let's summarize:

- Equilibrium in the goods market implies that an increase in the interest rate leads to a decrease in output. This relation is represented by the downward-sloping IS curve.
- Changes in factors that decrease the demand for goods, given the interest rate, shift the IS curve to the left. Changes in factors that increase the demand for goods, given the interest rate, shift the IS curve to the right.

5-2 FINANCIAL MARKETS AND THE LM RELATION

Let's now turn to financial markets. We saw in Chapter 4 that the interest rate is determined by the equality of the supply of and demand for money:

$$M = \$Y\, L(i)$$

The variable M on the left side is the nominal money stock. We shall ignore here the details of the money-supply process that we saw in Section 4-3, and simply think of the central bank as controlling M directly.

The right side gives the demand for money, which is a function of nominal income, $\$Y$, and of the nominal interest rate, i. As we saw in Section 4-1, an increase in nominal income increases the demand for money; an increase in the interest rate decreases the demand for money. Equilibrium requires that money supply (the left side of the equation) be equal to money demand (the right side of the equation).

Real Money, Real Income, and the Interest Rate

From Chapter 2: Nominal GDP = Real GDP multiplied by the GDP deflator $\$Y = YP$. Equivalently: ▶ Real GDP = Nominal GDP divided by the GDP deflator $\$Y/P = Y$.

The equation $M = \$Y\, L(i)$ gives a relation between money, nominal income, and the interest rate. It will be more convenient here to rewrite it as a relation among real money (that is, money in terms of goods), real income (that is, income in terms of goods), and the interest rate.

Recall that nominal income divided by the price level equals real income, Y. Dividing both sides of the equation by the price level P gives

$$\frac{M}{P} = Y\, L(i) \tag{5.3}$$

Hence, we can restate our equilibrium condition as the condition that the *real money supply*—that is, the money stock in terms of goods, not dollars—be equal to the *real money demand*, which depends on real income, Y, and the interest rate, i.

The notion of a "real" demand for money may feel a bit abstract, so an example will help. Instead of thinking of your demand for money in general, think of your demand for coins. Suppose you like to have coins in your pocket to buy two cups of coffee during the day. If a cup costs $1.20, you will want to keep about $2.40 in coins. This is your nominal demand for coins. Equivalently, you want to keep enough coins in your pocket to buy two cups of coffee. This is your demand for coins in terms of goods—here in terms of cups of coffee.

From now on, we shall refer to equation (5.3) as the LM *relation*. The advantage of writing things this way is that *real income, Y*, appears on the right side of the equation instead of *nominal income*, $\$Y$. And real income (equivalently real output) is the variable we focus on when looking at equilibrium in the goods market. To make the reading lighter, we will refer to the left and right sides of equation (5.3) simply as "money supply" and "money demand" rather than the more accurate but heavier "real money supply" and "real money demand." Similarly, we will refer to income rather than "real income."

Deriving the LM Curve

In deriving the IS curve, we took the two policy variables as government spending, G, and taxes, T. In deriving the LM curve—the curve corresponding to the LM relation—we have to decide how we characterize monetary policy: as the choice of M, the money stock, or the choice of i, the interest rate.

If we think of monetary policy as choosing the nominal money supply, M, and, by implication, given the price level that we shall take as fixed in the short run, choosing M/P, the real money stock, equation (5.3) tells us that real money demand, the right-hand side of the equation, must be equal to the *given* real money supply, the left-hand side of the equation. Thus, if, for example, real income increases, increasing money demand, the interest rate must increase so that money demand remains equal to the given money supply. In other words, for a given money supply, an increase in income automatically leads to an increase in the interest rate.

◄ Go back to Figure 4-4 in the previous chapter.

This is the traditional way of deriving the LM relation and the resulting LM curve. As we discussed in Chapter 4, however, the assumption that the central bank chooses the money stock and then just lets the interest rate adjust is at odds with reality today. Although, in the past, central banks thought of the money supply as the monetary policy variable, they now focus directly on the interest rate. They choose an interest rate, call it $\bar{i}$, and adjust the money supply so as to achieve it. Thus, in the rest of the book, we shall think of the central bank as choosing the interest rate (and doing what it needs to do with the money supply to achieve this interest rate). This will make for an extremely simple LM curve, namely, a horizontal line (in Figure 5-4) at the value of the interest rate, $\bar{i}$, chosen by the central bank.

◄ LM curve is a bit of a misnomer as, under our assumption, the LM relation is a simple horizontal line. But the use of the term *curve* is traditional, and I shall follow tradition.

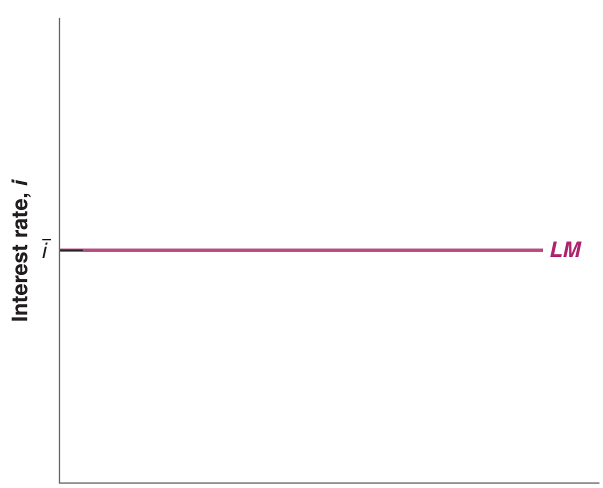


Figure 5-4

The LM Curve

The central bank chooses the interest rate (and adjusts the money supply so as to achieve it).

5-3 PUTTING THE IS AND LM RELATIONS TOGETHER

The IS relation follows from goods market equilibrium. The LM relation follows from financial market equilibrium. They must both hold.

$$\textit{IS relation}:\ Y = C(Y - T) + I(Y, i) + G$$
$$\textit{LM relation}:\ i = \bar{i}$$

Together they determine output. Figure 5-5 plots both the IS curve and the LM curve on one graph. Output—equivalently, production or income—is measured on the horizontal axis. The interest rate is measured on the vertical axis.

Any point on the downward-sloping IS curve corresponds to equilibrium in the goods market. *Any point* on the horizontal LM curve corresponds to equilibrium in financial markets. *Only at point A* are both equilibrium conditions satisfied. That means point A, with the associated level of output Y and interest rate $\bar{i}$, is the overall equilibrium—the point at which there is equilibrium in both the goods market and the financial markets.

You may ask: So what if the equilibrium is at point A, how does this fact translate into anything directly useful about the world? Don't despair: Figure 5-5 holds the answer to many questions in macroeconomics. The IS and LM relations that underlie the figure contain a lot of information about consumption, investment, and equilibrium conditions. Used properly, Figure 5-5 allows us to study what happens to output when the central bank decides to decrease the interest rate, or when the government decides to increase taxes, or when consumers become more pessimistic about the future, and so on.

In future chapters, you will see how we can extend it to think about the financial crisis, or about the role of expectations, or about the role of policy in an open economy.

Let's now see what the IS-LM model tells us, by looking separately at the effects of fiscal and monetary policy.

Fiscal Policy

Decrease in G-T ⇔ fiscal contraction ⇔ fiscal consolidation

Increase in G-T ⇔ fiscal expansion

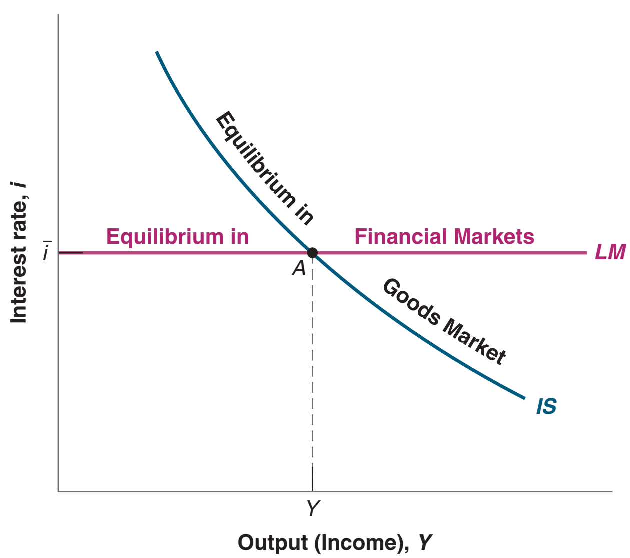
A reduction in the budget deficit, achieved either by increasing taxes, or by decreasing spending, or both, is called a **fiscal contraction** or a **fiscal consolidation**. Symmetrically, an increase in the budget deficit, achieved either by decreasing taxes, or by increasing spending, or both, is called a **fiscal expansion**.

Suppose the government decides to reduce the fiscal deficit, and to achieve this fiscal contraction through an increase in taxes. What will be the effects on output, on its composition, and on the interest rate?

Figure 5-5

The IS-LM Model

Equilibrium in the goods market implies that an increase in the interest rate leads to a decrease in output. This is represented by the IS curve. Equilibrium in financial markets is represented by the horizontal LM curve. Only at point *A*, which is on both curves, are both goods and financial markets in equilibrium.



When you answer this or any question about the effects of changes in policy (or, more generally, changes in exogenous variables), always go through the following three steps:

1. Ask how the change affects equilibrium in the goods market and how it affects equilibrium in the financial markets. Put another way: Does it shift the IS curve and/or the LM curve, and, if so, how?
2. Characterize the effects of these shifts on the intersection of the IS and LM curves. What does this do to equilibrium output and the equilibrium interest rate?
3. Describe the effects in words.

With time and experience, you will often be able to go directly to step 3. By then you will be ready to give an instant commentary on the economic events of the day. But until you get to that level of expertise, go step by step.

In this case, the three steps are easy. But going through them is good practice.

And when you feel really confident, put on a bow tie and go explain events on TV. (Why so many TV economists wear bow ties is a mystery.)

- Start with step 1. The first question is how the increase in taxes affects equilibrium in the goods market—that is, how it affects the relation between output and the interest rate captured in the IS curve. We derived the answer in Figure 5-3: At a given interest rate, the increase in taxes decreases output. The IS curve shifts to the left, from *IS* to *IS′* in Figure 5-6.

  Next, let's see if anything happens to the LM curve. By assumption, as we are looking at a change only in fiscal policy, the central bank does not change the interest rate. Thus, the LM curve, i.e., the horizontal line at $i = \bar{i}$, remains unchanged—it does not shift.
- Now consider step 2, the determination of the equilibrium.

  Before the increase in taxes, the equilibrium is given by point A, at the intersection of the *IS* and LM curves. After the increase in taxes and the shift to the left of the IS curve from *IS* to *IS′*, the new equilibrium is given by point A'. Output decreases from Y to Y'. By assumption, the interest rate does not change. Thus, as the IS curve *shifts*, the economy *moves along* the LM curve, from A to A'. The reason these words are italicized is that it is important always to distinguish between the *shift of* a curve (here the shift of the IS curve) and *movement along* a curve (here the movement along the LM curve). Many mistakes result from not distinguishing between the two.

The increase in taxes shifts the IS curve. The LM curve does not shift. The economy moves along the LM curve.

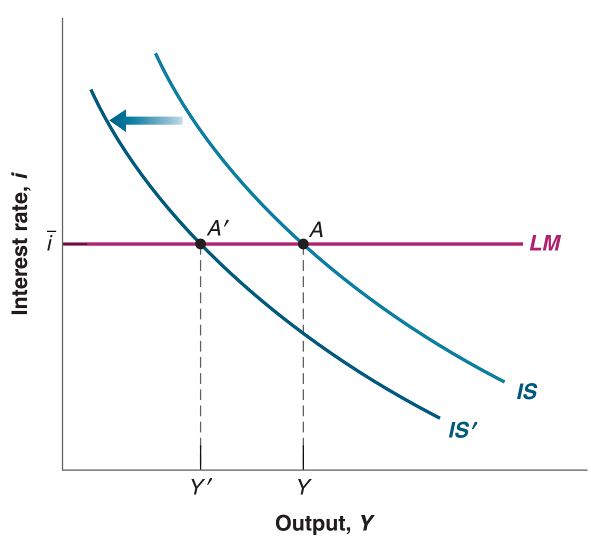


Figure 5-6

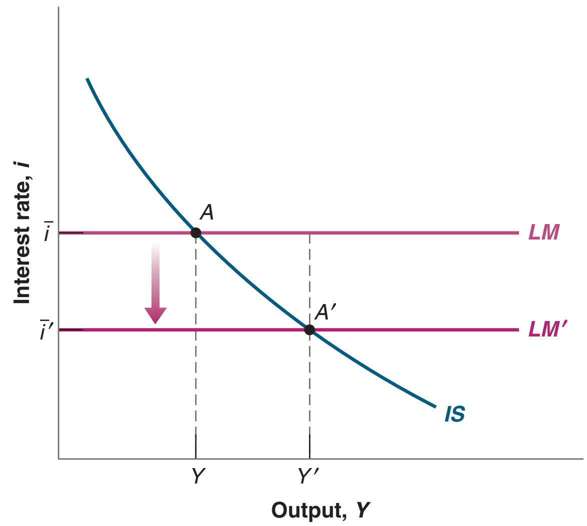
The Effects of an Increase in Taxes

An increase in taxes shifts the IS curve to the left. This leads to a decrease in the equilibrium level of output.

Figure 5-7

The Effects of a Decrease in the Interest Rate

A monetary expansion shifts the LM curve down and leads to higher output.



Note that we have just given a formal treatment of the informal discussion of the effects of an increase in public saving given in the Focus Box on "The Paradox of Saving" in Chapter 3.

- Step 3 is to tell the story in words:
  At a given interest rate, the increase in taxes leads to lower disposable income, which causes people to decrease their consumption. This decrease in demand leads, through a multiplier, to a decrease in output and income and, by implication, a decrease in investment.

Monetary Policy

Decrease in $i \Leftrightarrow$ increase in $M \Leftrightarrow$ monetary expansion.

Increase in $i \Leftrightarrow$ decrease in $M \Leftrightarrow$ monetary contraction (or monetary tightening).

Make sure you understand why the IS curve does not shift.

Now we turn to monetary policy. Suppose the central bank decreases the interest rate. Recall that, to do so, it increases the money supply. Such a change in monetary policy is called a **monetary expansion**. (Conversely, an increase in the interest rate, which is achieved through a decrease in the money supply, is called a **monetary contraction** or **monetary tightening**.)

- Again, step 1 is to see whether and how the IS and the LM curves shift.
  Let's look at the IS curve first. The change in the interest rate does not change the relation between output and the interest rate. It does not shift the IS curve.
  The change in the interest rate, however, leads (trivially) to a shift in the LM curve: it shifts down, from the horizontal line at $i = \bar{i}$ to the horizontal line $i = \bar{i}'$.
- Step 2 is to see how these shifts affect the equilibrium, represented in Figure 5-7. The economy moves down along the IS curve, and the equilibrium moves from point A to point A'. Output increases from Y to Y', and the interest rate decreases from i to i'.
- Step 3 is to say it in words: The lower interest rate leads to an increase in investment and, in turn, an increase in demand and output. Looking at the components of output: The increase in output and the decrease in the interest rate both lead to an increase in investment. The increase in income leads to an increase in disposable income and, in turn, consumption, so both consumption and investment increase.

5-4 USING A POLICY MIX

We have looked so far at fiscal policy and monetary policy in isolation. Our purpose was to show how each worked. In practice, the two are often used together. The combination of monetary and fiscal policies is known as the **monetary-fiscal policy mix**, or simply the *policy mix*.

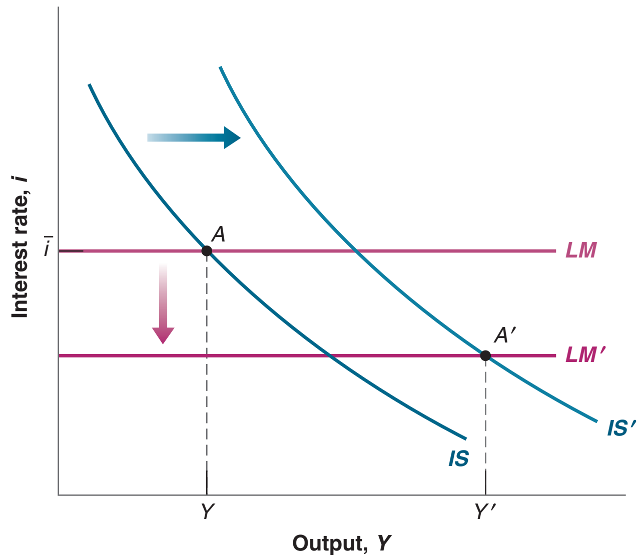


Figure 5-8

The Effects of a Combined Fiscal and Monetary Expansion

Fiscal expansion shifts the IS curve to the right. Monetary expansion shifts the LM curve down. Both lead to higher output.

Sometimes, the right mix is to use fiscal and monetary policy in the same direction. Suppose, for example, that the economy is in a recession and output is too low. Then, both fiscal and monetary policies can be used to increase output. This combination is represented in Figure 5-8. The initial equilibrium is given by the intersection of *IS* and *LM* at point *A*, with corresponding output *Y*. Expansionary fiscal policy, say through a decrease in taxes, shifts the IS curve to the right, from *IS* to *IS*′. Expansionary monetary policy shifts the LM curve down from *LM* to *LM*′. The new equilibrium is at *A*′, with corresponding output *Y*′. Thus, both fiscal and monetary policies contribute to the increase in output: Lower taxes and higher income result in higher consumption, which in turn leads to higher output and, together with a lower interest rate, higher investment.

Such a combination of fiscal and monetary policy is typically used to fight recessions. An example is given in the Focus Box "The European Sovereign Debt Crisis with a Special Focus on Greece during 2010–2013."

We shall see other examples later in the book. Section 6-5 looks at the role of fiscal and monetary policy during the Great Financial Crisis.

You might ask: Why use both policies when either one on its own could achieve the desired increase in output? An increase in output could in principle be achieved just by using fiscal policy—say through a sufficiently large increase in government spending, or a sufficiently large decrease in taxes—or just by using monetary policy, through a sufficiently large decrease in the interest rate. The answer is that there are a number of reasons why policymakers may want to use a policy mix.

- A fiscal expansion means either an increase in government spending, or a decrease in taxes, or both. This means an increase in the budget deficit (or, if the budget was initially in surplus, a smaller surplus). As we shall see later, but you surely can guess why already, running a large deficit and increasing government debt could be dangerous later. In this case, it is better to rely, at least in part, on monetary policy.

More on this in Chapter 22.

- A monetary expansion means a decrease in the interest rate. If the interest rate is very low, then the room for using monetary policy may be limited. In this case, fiscal policy has to do more of the job. If the interest rate is already equal to zero because of *the zero lower bound*, fiscal policy has to do all the work. As we saw in Chapter 1, while US interest rates have become positive, they remain low. If demand were to decrease soon, the room for monetary policy to decrease interest rates would be limited, and fiscal policy would have to play the main role.

FOCUS

The European Sovereign Debt Crisis with a Special Focus on Greece during 2010–2013

The foundation of euro was laid in 1992, when the European Union was established by the Maastricht Treaty. To join the currency union, member states had to qualify by meeting the terms of the Maastricht Treaty in terms of budget deficits, inflation, interest rates, and other fiscal requirements. In 2001, Greece joined the euro. As discussed in Chapter 1, in the aftermath of the Great Financial Crisis, the enormous increase in sovereign debt emerged as an important negative outcome, since public debt was dramatically increased in an effort by the European governments (following the US) to reduce the accumulated growth of private debt in the years preceding the recent financial turmoil. While other advanced economies slowly recovered from the crisis, the euro area faced a recession that challenged the viability of the currency union. So, what triggered the euro area recession? The causes of the crisis can generally be pinned to three broad underlying factors—policy failures that allowed imbalances to increase, insufficient institutions in the euro area to absorb shocks, and crisis mismanagement. In the series of events, Greece was the first domino to fall in Europe.

On the fiscal policy front, Greece slashed spending and raised taxes to fight the crisis as public deficits reached very high levels: −6.4% in 2009 and −6.2% in 2010, both expressed as ratios to GDP.

In September 2009, with an economy weakened by reduced tax revenues and increased spending demands, the Greek government projected a budget deficit of 6% for the year and a debt-to-GDP ratio of nearly 100%. However, in 2010, Eurostat revealed that the newly elected Greek government has announced how the previous governments had undervalued the true size of the country's budget deficit—a whopping 15.4%—and stated that government debt actually amounted to 126.8%, which was 11 percentage points higher than what had been previously reported. Greece tried to save itself but failed. It was caught in a public debt vortex. Meanwhile, the austerity measures backfired and fanned the recession by reducing tax revenues and increasing social spending, pushing the nation closer to the precipice of unsustainability. In December 2009, Greece admitted that its debts reached €300 billion, the highest in modern history. Credit-rating agencies repeatedly downgraded the Greek government debt and its borrowing cost rapidly rose from 1.5% to 5%.

The weakness of the institutional framework became clear with the threat of the Greek default affecting other weak European economies, like Italy, Portugal, Spain, and Cyprus, making the financial community (**banks and investors**) realize that a common currency did not mean that Europe was a single entity when it came to credit risk.

In terms of monetary policy, in May 2009, right at the beginning of the crisis, the ECB lowered its benchmark interest rate to 1% and limited its open market operations to €1,000 billion in February 2010 and December 2011. In 2010, Greece's 2009 budget deficit was revised upward to 12.7%, from 3.7%, and the euro continued to fall against the dollar and the pound. In the same year, a "troika" was formed with the IMF, the European Commission, and the ECB, and the European Financial Stabilization Fund (EFSF), a temporary emergency resolution mechanism, was created to provide financial support to Greece, Ireland, and Portugal—€68 billion for Ireland, €78 billion for Portugal, and more than €320 billion for Greece—contingent upon the countries setting up austerity programs. These measures, however, were not sufficient for banks to overcome liquidity and solvency problems.

In 2011, the volatile economic situation continued to expand, plunging the entire euro area into recession in 2012 and 2013. Despite austerity measures aimed at dramatically cutting government spending and raising taxes, interest rates on Greek debt soared, eventually rising to nearly 40%, and the debt-to-GDP ratio climbed to 160% in 2012. Even with bailouts from other European countries and liquidity support from the European Central Bank, Greece was forced to write down the value of its debt held in private hands by more than half, and the country was subject to civil unrest, with massive strikes and the resignation of the prime minister.

The risk of default on government debt had hit at a time when European banks considered all European government bonds to be similar, bearing almost no risks. Thus, most banks held large amounts of euro area sovereign debt in their balance sheets. In addition, the aftermath of the Great Financial Crisis cast doubts on mortgage loan assets of several banks. This resulted in a blockage of interbank credit, a fragmentation of the financial European system, and a risk of general collapse of the euro system.

Many economists believe that the European governments not only mismanaged the crisis but also aggravated the situation by continuing to stick to the Union's fiscal parameters of debt and deficit instead of devising policy measures tailored to the gravity of the situation. The consequence of this was that they did not utilize fiscal policy measures to balance the recession. Being independent of any government, the European Central Bank (ECB) could not finance the fiscal deficit of any EU member country. Where the Federal Reserve Bank was working on creating maximum sustainable employment and stable prices, the ECB was focusing only on price stability, which meant keeping inflation below but close to 2%.

In July 2012, the markets began to calm down after Mario Draghi, the president of the European Central Bank, said that the ECB was ready to do "whatever it takes" to save the euro. However, it wasn't until 2015 that the ECB decided to implement an open-ended quantitative easing program and purchase the sovereign debt bonds detained by the commercial banks. As confidence levels gradually recovered, the euro area began seeing slow but positive economic growth.

- Fiscal and monetary policies have different effects on the composition of output. A decrease in income taxes, for example, tends to increase consumption relative to investment. A decrease in the interest rate affects investment more than consumption. Thus, depending on the initial composition of output, policymakers may want to rely more on fiscal or more on monetary policy.
- Finally, neither fiscal policy nor monetary policy works perfectly. A decrease in taxes may fail to increase consumption. A decrease in the interest rate may fail to increase investment. Thus, in case one policy does not work as well as hoped for, it is better to use both.

Sometimes, the right policy mix is instead to use the two policies in opposite directions, for example, combining a fiscal consolidation with a monetary expansion. Suppose, for example, that the government is running a large budget deficit and would like to reduce it, but does not want to trigger a recession. In Figure 5-9, the initial equilibrium is given by the intersection of the IS and LM curves at point A, with associated output Y. Output is thought to be at the right level, but the budget deficit, T-G, is too large.

If the government reduces the deficit, say by increasing T or by decreasing G (or both), the IS curve will shift to the left, from IS to IS'. The equilibrium will be at point A', with level of output Y'. At a given interest rate, higher taxes or lower spending will decrease demand, and through the multiplier, decrease output. Thus, the reduction in the deficit will lead to a recession.

The recession can be avoided, however, if monetary policy is also used. If the central bank reduces the interest rate to $\bar{i}'$, the equilibrium is given by point A'' with corresponding output $Y'' = Y$. The combination of both policies thus allows for the reduction in the deficit, but without a recession.

What happens to consumption and investment in this case? What happens to consumption depends on how the deficit is reduced. If the reduction takes the form of a decrease in government spending rather than an increase in taxes, income is unchanged, disposable income is unchanged, and so consumption is unchanged. If the reduction takes the form of an increase in income taxes, then disposable income is lower, and so is consumption. What happens to investment is unambiguous: Unchanged output and a lower interest rate imply higher investment. The relation between deficit reduction and investment is discussed further in the Focus Box "Deficit Reduction: Good or Bad for Investment?"

Such a policy mix was used in the early 1990s. When Bill Clinton was elected president in 1992, one of his priorities was to reduce the budget deficit using a combination

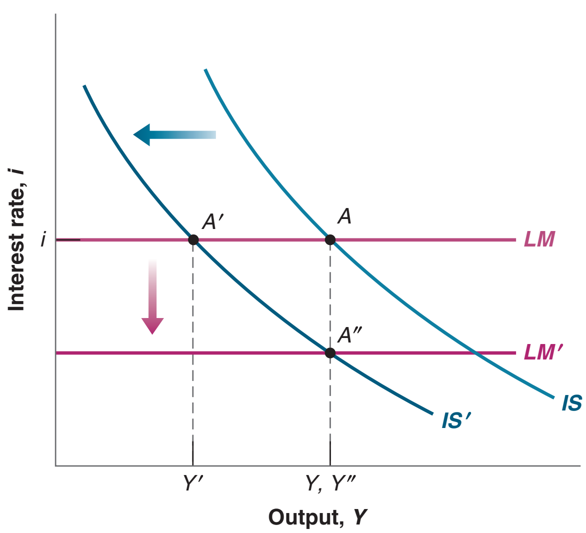


Figure 5-9

The Effects of a Combined Fiscal Consolidation and a Monetary Expansion

The fiscal consolidation shifts the IS curve to the left. A monetary expansion shifts the LM curve down. Together, they leave output unchanged while the budget deficit is reduced.

FOCUS

Deficit Reduction: Good or Bad for Investment?

You may have heard this argument in some form: "Private saving goes either toward financing the budget deficit or financing investment. It does not take a genius to conclude that reducing the budget deficit leaves more saving available for investment, so investment increases."

This argument sounds convincing. But, as we have seen earlier in the book, it must be wrong. If, for example, deficit reduction is not accompanied by a decrease in the interest rate, then we know that output decreases (see Figure 5-7) and, by implication, so does investment since it depends on output. So what is going on in this case?

Go back to Chapter 3, equation (3.10). There we learned that we can also think of the goods-market equilibrium condition as

$$\begin{array}{ccccc} \text{Investment} & = & \text{Private saving} & + & \text{Public saving} \\ I & = & S & + & (T - G) \end{array}$$

In equilibrium, investment is equal to private saving plus public saving. If public saving is positive, the government is said to be running a budget surplus; if public saving is negative, the government is said to be running a budget deficit. So it is true that *given private saving*, if the government reduces its deficit—either by increasing taxes or reducing government spending so that T-G goes up—investment must go up: Given S, T-G going up implies that I goes up.

The crucial part of this statement, however, is "given private saving." The point is that a fiscal contraction affects private saving as well: The contraction leads to lower output and therefore lower income. As consumption goes down by less than income, private saving also goes down. It actually goes down by more than the reduction in the budget deficit, leading to a decrease in investment. In terms of the equation: S decreases by more than T-G increases, and so I decreases. (You may want to do the algebra and convince yourself that saving actually goes down by *more* than the increase in T-G.)

Does this mean that deficit reduction always decreases investment? The answer is clearly *no*. We saw this in Figure 5-9. If, when the deficit is reduced, the central bank also decreases the interest rate so as to keep output constant, then investment necessarily goes up. Although output is unchanged, the lower interest rate leads to higher investment.

The moral of this box is clear: Whether deficit reduction leads to an increase in investment is far from automatic. It may or it may not, depending on the response of monetary policy.

of cuts in spending and increases in taxes. He was worried, however, that, by itself, such a fiscal contraction would lead to a decrease in demand and trigger another recession. The right strategy was to combine a fiscal contraction (to get rid of the deficit) with a monetary expansion (to make sure that demand and output remained high). This was the strategy adopted and carried out by Clinton (who was in charge of fiscal policy) and Alan Greenspan (who was in charge of monetary policy). The result of this strategy—and a bit of economic luck—was a steady reduction of the budget deficit (which turned into a budget surplus at the end of the 1990s) and a steady increase in output throughout the rest of the decade.

A similar discussion is taking place today in the euro area but with a twist. Worried about high public debt levels, governments would like to reduce fiscal deficits to decrease debt levels over time, a policy known as **fiscal austerity**. The problem, however, is that interest rates are already very low, so there is little room for monetary policy to offset the adverse effects of a fiscal contraction on output. This is leading to an intense debate between those who think that fiscal austerity is needed, even if it has adverse effects on output, and those who think that fiscal consolidation should wait until monetary policy can be used to offset its adverse effect on output.

5-5 HOW DOES THE IS-LM MODEL FIT THE FACTS?

We have so far ignored dynamics. For example, when looking at the effects of an increase in taxes in Figure 5-6—or the effects of a monetary expansion in Figure 5-7—we made it look as if the economy moved instantaneously from A to A', as if output went instantaneously from Y to Y'. This is clearly not realistic: The adjustment of output takes time. To capture this time dimension, we need to reintroduce dynamics.

Introducing dynamics formally would be difficult. But, as we did in Chapter 3, we can describe the basic mechanisms in words. Some of the mechanisms will be familiar from Chapter 3, some are new:

- Consumers are likely to take some time to adjust their consumption following a change in disposable income.
- Firms are likely to take some time to adjust investment spending following a change in their sales.
- Firms are likely to take some time to adjust investment spending following a change in the interest rate.
- Firms are likely to take some time to adjust production following a change in their sales.

So, in response to an increase in taxes, it takes some time for consumption spending to respond to the decrease in disposable income, some more time for production to decrease in response to the decrease in consumption spending, yet more time for investment to decrease in response to lower sales, and so on.

In response to the decrease in the interest rate, it takes some time for investment spending to respond, some more time for production to increase in response to the increase in demand, yet more time for consumption and investment to increase in response to the induced change in output, and so on.

Describing precisely the adjustment process implied by all these sources of dynamics is complicated, but the basic implication is straightforward: Time is needed for output to adjust to changes in fiscal and monetary policy. How much time? This question can be answered only by looking at the data and using econometrics. Figure 5-10 shows the results of such an econometric study, which uses data from the United States from 1960 to 1990.

The study looks at the effects of a decision by the Fed to increase the federal funds rate by 1%. It traces the typical effects of such an increase on a number of macroeconomic variables.

We discussed the federal funds market and the federal funds rate in Chapter 4, Section 4–3.

Each panel in Figure 5-10 represents the effects of the change in the interest rate on a given variable: retail sales, output, employment, unemployment, and prices. Each panel plots three lines: the solid line in the center of a band gives the best estimate of the effect of the interest rate change on the variable, and the two dashed lines and tinted space between them represent a **confidence band**, within which the true value of the effect lies with 60% probability.

There is no such thing in econometrics as learning the exact value of a coefficient or the exact effect of one variable on another. Rather, econometrics provides a best estimate (here, the solid line) and a measure of confidence in the estimate (the confidence band).

- Figure 5-10(a) shows the effects of an increase in the federal funds rate of 1% on retail sales over time (12 quarters). The percentage change in retail sales is plotted on the vertical axis; time, measured in quarters, is on the horizontal axis.
  Focusing on the best estimate—the solid line—we see that the increase in the federal funds rate of 1% leads to a decline in retail sales. The largest decrease in retail sales, −0.9%, occurs after five quarters.
- Figure 5-10(b) shows that lower sales lead to lower output. In response to the decrease in sales, firms cut production, but by less than the decrease in sales. This is because firms accumulate inventories for some time, so the adjustment of production is smoother and slower than the adjustment of sales. The largest decrease, −0.7%, is reached after eight quarters.

In other words, monetary policy works, but with long lags. It takes nearly two years for monetary policy to have its full effect on output.

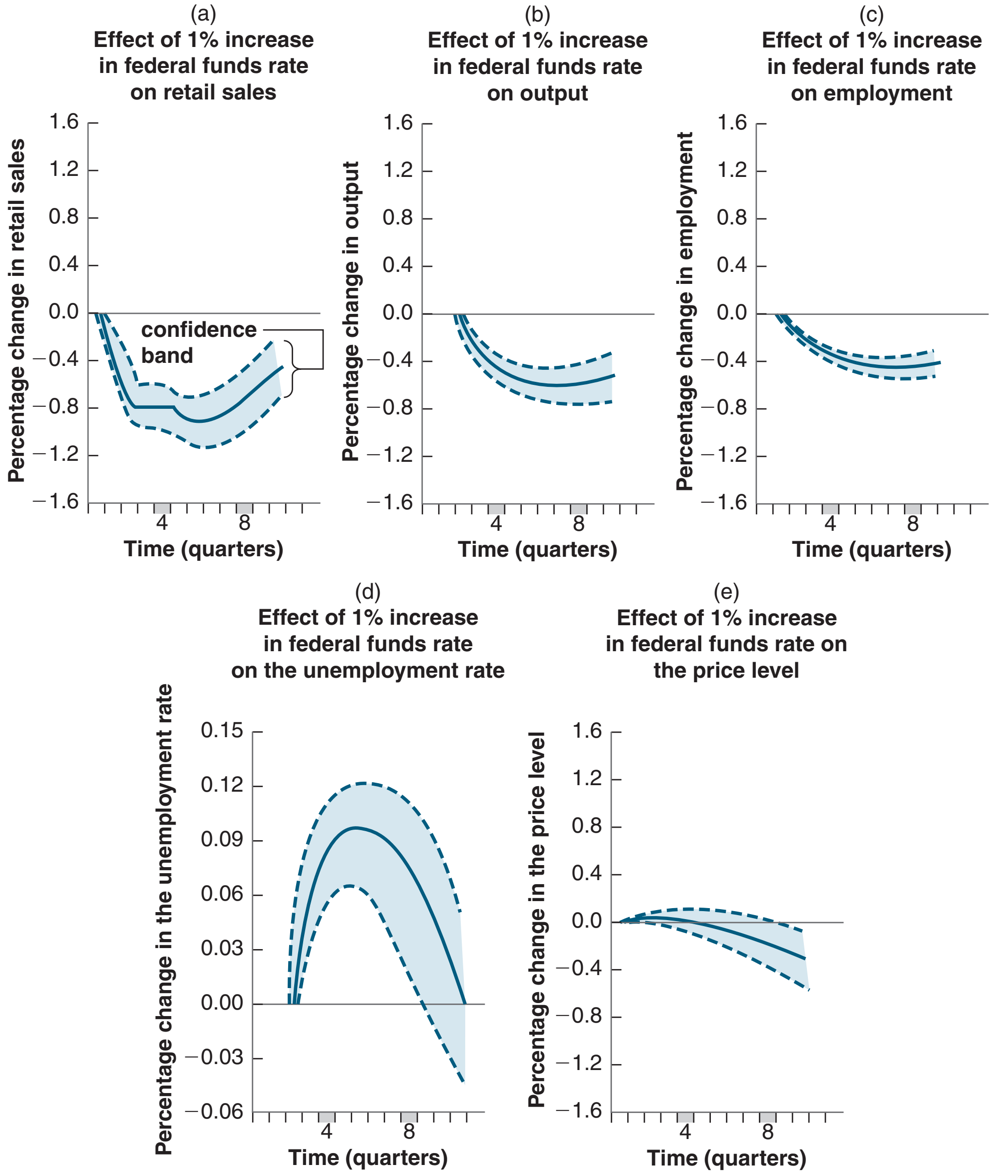
This explains why monetary policy could not have prevented the 2001 recession. When the Fed started decreasing the federal funds rate at the start of 2001, it was already too late for these cuts to have much effect that year.

Figure 5-10

The Empirical Effects of an Increase in the Federal Funds Rate

In the short run, an increase in the federal funds rate leads to a decrease in sales, output, and employment and an increase in unemployment, but it has little effect on the price level.

Source: Lawrence Christiano, Martin Eichenbaum, and Charles Evans, "The Effects of Monetary Policy Shocks: Evidence from the Flow of Funds," *Review of Economics and Statistics*, 1996, 78 (February): pp. 16–34.



- Figure 5-10(c) shows that lower output leads to lower employment: As firms cut production, they also cut employment. As with output, the decline in employment is slow and steady, reaching −0.5% after eight quarters. The decline in employment leads to an increase in the unemployment rate, shown in Figure 5-10(d).
- Figure 5-10(e) looks at the behavior of the price level. Remember, one of the *assumptions* of the IS-LM model is that the price level is given, and so it does not change in response to changes in demand. Figure 5-10(e) shows that this assumption is not a bad approximation of reality in the short run. The price level is nearly unchanged for the first six quarters or so. Only after the first six quarters does it decline. This gives us a strong hint as to why the IS-LM model becomes less reliable as we look at the medium run: In the medium run, we can no longer assume that the price level is given, and movements in the price level become important.

Figure 5-10 provides two important lessons. First, it gives a sense of the dynamic adjustment of output and other variables to monetary policy.

Second, and more fundamentally, it shows that what we observe in the economy is consistent with the implications of the IS-LM model. This does not *prove* that the IS-LM model is the right model. It may be that what we observe in the economy is the result of a completely different mechanism, and the fact that the IS-LM model fits well is a coincidence. But this seems unlikely. The IS-LM model looks like a solid basis on which to build when looking at movements in activity in the short run. Later on, we shall extend the model to look at the role of expectations (Chapters 14 to 16) and the implications of openness in goods and financial markets (Chapters 17 to 20). But we must first understand what determines output in the medium run. This is the topic of Chapters 7 to 9.

SUMMARY

- The IS-LM model characterizes the implications of equilibrium in both the goods and financial markets.
- The IS relation and the IS curve show the combinations of the interest rate and the level of output that are consistent with equilibrium in the goods market. An increase in the interest rate leads to a decline in output. Consequently, the IS curve is downward sloping.
- The LM relation and the LM curve show the combinations of the interest rate and the level of output consistent with equilibrium in financial markets. Under the assumption that the central bank chooses the interest rate, the LM curve is a horizontal line at the interest rate chosen by the central bank.
- A fiscal expansion shifts the IS curve to the right, leading to an increase in output. A fiscal contraction shifts the IS curve to the left, leading to a decrease in output.
- A monetary expansion shifts the LM curve down, leading to a decrease in the interest rate and an increase in output. A monetary contraction shifts the LM curve up, leading to an increase in the interest rate and a decrease in output.
- The combination of monetary and fiscal policies is known as the monetary-fiscal policy mix, or simply the policy mix. Sometimes monetary and fiscal policy are used in the same direction, sometimes in opposite directions. Together, a fiscal contraction and a monetary expansion can, for example, achieve a decrease in the budget deficit while avoiding a decrease in output.
- The IS-LM model appears to describe well the behavior of the economy in the short run. In particular, the effects of monetary policy are similar to those implied by the IS-LM model once dynamics are introduced in the model. An increase in the interest rate due to a monetary contraction leads to a decrease in output, with the maximum effect taking place after about eight quarters.

KEY TERMS

IS curve, 111
LM curve, 113
fiscal contraction, 114
fiscal consolidation, 114
fiscal expansion, 114
monetary expansion, 116
monetary contraction, 116
monetary tightening, 116
monetary-fiscal policy mix, 116
fiscal austerity, 120
confidence band, 121

QUESTIONS AND PROBLEMS

QUICK CHECK

1. Using the information in this chapter, label each of the following statements true, false, or uncertain. Explain briefly.

a. The main determinants of investment are the level of sales and the interest rate.

b. If all the exogenous variables in the IS relation are constant, then a higher level of output can be achieved only by lowering the interest rate.

c. The IS curve is downward sloping because goods market equilibrium implies that an increase in taxes leads to a lower level of output.

d. If government spending and taxes increase by the same amount, the IS curve does not shift.

e. The LM curve is horizontal at the central bank's policy choice of the interest rate.

f. The real money supply is constant along the LM curve.

g. If the nominal money supply is $400 billion and the price level rises from an index value of 100 to an index value of 103, the real money supply rises.
h. If the nominal money supply rises from $400 billion to $420 billion and the price level rises from an index value of 100 to 102, the real money supply rises.
i. An increase in government spending leads to a decrease in investment in the IS-LM model.

2. Consider first the goods market model with constant investment that we saw in Chapter 3. Consumption is given by

$$C = c_0 + c_1(Y - T)$$

and I, G, and T are given.

a. Solve for equilibrium output. What is the value of the multiplier for a change in autonomous spending?
Now let investment depend on both sales and the interest rate:

$$I = b_0 + b_1 Y - b_2 i$$

b. Solve for equilibrium output using the methods learned in Chapter 3. At a given interest rate, why is the effect of a change in autonomous spending bigger than what it was in part a? (Assume $c_1 + b_1 < 1$.)
c. Suppose the central bank chooses an interest rate of $\bar{i}$. Solve for equilibrium output at that interest rate.
d. Draw the equilibrium of this economy using an IS-LM diagram.

3. The response of the economy to fiscal policy
a. Use an IS-LM diagram to show the effects on output of a decrease in government spending. Can you tell what happens to investment? Why?
Now consider the following IS-LM model:

$$C = c_0 + c_1(Y - T)$$
$$I = b_0 + b_1 Y - b_2 i$$
$$Z = C + I + G$$
$$i = \bar{i}$$

b. Solve for equilibrium output when the interest rate is $\bar{i}$. Assume $c_1 + b_1 < 1$. (*Hint*: You may want to rework through Problem 2 if you are having trouble with this step.)
c. Solve for equilibrium level of investment.
d. Let's go behind the scenes in the money market. Chapter 4 introduced equations that describe equilibrium in the money market. Let's write the equation characterizing the equilibrium as: $M/P = d_1 Y - d_2 i$. Solve for the equilibrium level of the real money supply when $i = \bar{i}$. How does the real money supply vary with government spending?

4. Consider the money market to better understand the horizontal LM curve in this chapter.

The LM relation (equation 5.3) is $\frac{M}{P} = Y\,L(i)$

a. What is on the left-hand side of equation (5.3)?
b. What is on the right-hand side of equation (5.3)?
c. Go back to Figure 4-2 in the previous chapter. How is the function $L(i)$ represented in that figure?
d. Modify Figure 4-2 to represent equation (5.3) in two ways. How does the horizontal axis have to be relabeled? What is the variable that now shifts the money demand function? Draw a modified Figure 4-2 with the appropriate labels.
e. Use your modified Figure 4-2 to show that (1) as output rises, to keep the interest rate constant, the central bank must increase the real money supply; (2) as output falls, to keep the interest rate constant, the central bank must decrease the real money supply.

5. Consider the following numerical example of the IS-LM model:

$$C = 100 + 0.3Y_D$$
$$I = 150 + 0.2Y - 1000i$$
$$G = 200$$
$$T = 100$$
$$\bar{i} = 0.03$$

a. Derive the IS relation. (*Hint*: You want an equation with Y on the left side and everything else on the right.)
b. The central bank sets an interest rate of 3%. How is that decision represented in the equations?
c. What is the level of real money supply when the interest rate is 3%? Use the expression:

$$(M/P) = 2Y - 4000i$$

d. Solve for the equilibrium values of C and I, and verify the value you obtained for Y by adding C, I, and G.
e. Now suppose that the central bank increases money supply to 1500. How does this change the LM curve? Solve for Y, I, and C, and describe in words the effects of an expansionary monetary policy.
f. Return to the initial situation in which the interest rate set by the central bank is 3%. Suppose that government spending increases to $G = 300$. Summarize the effects of an expansionary fiscal policy on Y, I, and C. What is the effect of the expansionary fiscal policy on the real money supply?

DIG DEEPER

6. Investment and the interest rate

The chapter argues that investment depends negatively on the interest rate because an increase in the cost of borrowing discourages investment. However, firms often finance their investment projects using their own funds.

If a firm is considering using its own funds (rather than borrowing) to finance investment projects, will higher interest rates discourage the firm from undertaking these projects? Explain. (*Hint*: Think of yourself as the owner of a firm that has earned profits and imagine that you are going to use the profits either to finance new investment projects or to buy bonds. Will your decision to invest in new projects in your firm be affected by the interest rate?)

7. The fiscal-monetary policy mix in the aftermath of the Great Financial Crisis

The Great Financial Crisis left many nations with slow GDP growth rates and high levels of public debt. While most nations pursued a monetary policy, some nations simply lowered income taxes through expansionary fiscal policy.

a. Illustrate the effect of such a policy mix on output.
b. Why do the policy mix differ for different nations in 2008?
c. Explain if these policy mixes were successful in combatting the global meltdown.

8. *What mix of monetary and fiscal policy is needed to meet the following objectives?*

a. Increase Y while keeping $\bar{i}$ constant. Would investment (I) change?
b. Decrease a fiscal deficit while keeping Y constant. Why must $\bar{i}$ also change?

9. *The (less paradoxical) paradox of saving*

A chapter problem at the end of Chapter 3 considered the effect of a drop in consumer confidence on private saving and investment, when investment depended on output but not on the interest rate. Here, we consider the same experiment in the context of the IS-LM framework, in which investment depends on the interest rate and output but the central bank moves interest rates to keep output constant.

a. Suppose consumer confidence falls, so households save a higher proportion of their income. In an IS-LM diagram where the central bank moves interest rates to keep output constant, show the effect of the fall in consumer confidence on the equilibrium in the economy.
b. How will the fall in consumer confidence affect consumption, investment, and private saving? Will the attempt to save more necessarily lead to more saving? Will this attempt necessarily lead to less saving?

10. *Fiscal policy and investment. Read Focus Box "Deficit Reduction: Good or Bad for Investment?"*

In each case below, there is a fiscal consolidation. Remember that equilibrium condition in good markets can also be written

$$I = S + (T - G)$$

a. How does this fiscal consolidation increase public saving? Calculate the change in public saving and the change in private saving? What must happen to the target interest rate for this policy to describe the changes in parts a, b, and c?

Year	Y	C	I	G	T	S
Pre-policy	1000	500	200	300	200	
Post-policy	950	400	300	250	250	

b.

Year	Y	C	I	G	T	S
Pre-policy	1000	500	200	300	200	
Post-policy	900	450	250	200	200	

c.

Year	Y	C	I	G	T	S
Pre-policy	1000	500	200	300	200	
Post-policy	975	480	195	300	300	

EXPLORE FURTHER

11. *The Clinton-Greenspan policy mix*

As described in this chapter, during the Clinton administration the policy mix changed toward more contractionary fiscal policy and more expansionary monetary policy. This question explores the implications of this policy mix, in both theory and fact.

a. What must the Federal Reserve do to ensure that, if G falls and T rises, that combination of policies has no effect on output? Show the effects of these policies in an IS-LM diagram. What happens to the interest rate? What happens to investment?
b. Go to the website of the *Economic Report of the President* www.govinfo.gov/app/collection/erp/2019 and look at Table B-46 in the statistical appendix. What happened to federal receipts (tax revenues), federal outlays, and the budget deficit as a percentage of GDP over the period 1992 to 2000? (Note that federal outlays include transfer payments, which would be excluded from the variable G, as we define it in our IS-LM model. Ignore the difference.)
c. The Federal Reserve Board of Governors posts the recent history of the federal funds rate at www.federalreserve.gov/releases/h15/data.htm. You can look at the rate on a daily, weekly, monthly, or annual interval. Look at the years between 1992 and 2000. When did monetary policy become more expansionary?
d. Go to Table B-2 of the *Economic Report of the President* and collect data on real GDP and real gross domestic investment for the period 1992 to 2000. Calculate investment as a percentage of GDP for each year. What happened to investment over the period?
e. Finally, go to Table B-31 and retrieve data on real GDP per capita (in chained 2005 dollars) for the period. Calculate the growth rate for each year. What was the average annual growth rate over the period 1992 to 2000? In Chapter 10 you will learn that the average annual growth rate of US real GDP per capita was 2.6% between 1950 and 2004. How did growth between 1992 and 2000 compare to the post–World War II average?

12. *Consumption, investment, and GDP in China*

To answer this question, examine the changes in the components of GDP over this period and the movements of investment and consumption in China during the last two or three decades and its relative slowdown since the Great Financial Crisis.

Along with other emerging market economies, China has been considered as one of the main drivers of global growth especially *during the crisis. Since 1990, China has been able to sustain the highest levels of GDP growth, rendering it one of the largest economies in the world, second to the United States. But why has China's GDP growth been slower after the crisis?*

Visit the website of the National Bureau of Statistics of China at www.stats.gov.cn (click on English to translate the page from Mandarin) and to the IMF Article IV consultation on China (https://www.imf.org/en/Publications/CR/Issues/2017/08/15/People-s-Republic-of-China-2017-Article-IV-Consultation-Press-Release-Staff-Report-and-45170). *Find the annual National Accounts tables and examine the composition of GDP, which shows its percentage contribution to the overall growth.*

a. Track the change in GDP variable since 1990. Which variable had the bigger percentage change around this time? How can these changes explain the pattern of changes in China's economy? How has this pattern led to higher GDP growth from 1990 till 2010?

b. Why has this trend been reversed since the global financial crisis? To answer this question, you need to examine the levels of Chinese exports. Also, examine the yuan exchange rate fluctuation between 2010 and 2017, and analyze what could be its relationship with economic growth in China?

c. Plot the changes in investment (gross capital formation) and final consumption expenditure since 1990. Which variable had the bigger percentage change since 1990?

d. Which region in China contributes most to investment expenditure, final consumption expenditure, and GDP growth? How can you explain these changes? Research the factors that have enabled each of these regions to achieve these levels of consumption and investment.

FURTHER READING

- A description of the US economy, from the period of "irrational exuberance" to the 2001 recession, and the role of fiscal and monetary policy is given by Paul Krugman in *The Great Unraveling* (New York: W. W. Norton, 2003). (Warning: Krugman did not like the Bush administration or its policies!)

Low rates are increasingly seen as a permanent feature of the global economic landscape

My column this week sets out three possible scenarios for the American economy in 2016, in the aftermath of the Fed's first rate hike in more than nine years. Each scenario corresponds to an understanding of why it is that near-zero interest rates are so difficult to leave behind; economies eventually managed the trick in the decades after the Depression, but those that have sunk to the zero lower bound in recent years have been unable to escape it for long.

What strikes me as interesting, and what motivated the column, is that our understanding of the pull of near-zero rates has evolved since late 2008, and continues to evolve, in a very ominous direction.

Back in late 2008 and early 2009, when rates around the rich world fell below 1%, the framework most economists reached for was what you might call the traditional Hicks-Krugman story of the liquidity trap. John Hicks's analysis of the work of John Maynard Keynes first set out the concept of a liquidity trap in 1937. Paul Krugman borrowed and updated that framework in 1998 in an analysis of the Japanese economy. This story is one in which a really nasty economic shock knocks an economy into a bad equilibrium; rates fall to zero, at which point monetary policy loses its punch. Real rates can't go low enough to stimulate the economy, which remains stuck with a shortfall in demand. To get out, the government either needs to borrow heavily and spend to boost demand, or the central bank needs to promise to tolerate high inflation once, at some point in the distant future, the economy returns to health: to "credibly promise to be irresponsible", in Mr Krugman's phrase. Higher expected inflation reduces the real interest rate in the present, providing the needed stimulative jolt. In his paper, Mr Krugman mused that a target of 4% inflation for fifteen years might be the sort of thing needed to get Japan out of its trap – assuming Japanese households would find such a target credible.

An alternative view emerged over the course of the recession and recovery, which one might call the Friedman-Schwartz-Bernanke story. In the Monetary History of the United States, Milton Friedman and Anna Schwartz argued that monetary policy had not been helpless in the 1930s, and that in fact the blame for the depth and length of the Depression should be set at the feet of the Federal Reserve, which tolerated a dramatic drop in the money supply. Ben Bernanke's Fed adopted a version of this framework, which continues to shape policy today: that a liquidity trap is only a trap for an insufficiently aggressive central bank. Use enough unconventional monetary policy, and the trap can be overcome. And so the Fed never attempted to gin up any sort of regime change, or to dramatically increase the market's expectations for future inflation. Instead, it used QE and promises to keep rates low for as long as necessary to support demand. And the Fed now seems confident that, having generated a robust-enough recovery, it is safe to move away from zero, as nonchalantly as if one were raising rates from 4% to 4.25%.

That view is almost certainly wrong. Other rich-world central banks with other robust-enough recoveries have tried and failed to sidle away from zero; it doesn't work. Markets don't think it will work this time; futures markets project a path for the federal funds rate about half as steep as the (already gentle) path that the Fed suggests it will follow.

At the same time, the traditional liquidity trap story also looks inadequate. America has enjoyed a relatively robust recovery, at least over the last year or two, despite big government budget cuts and inflation rates, both actual and expected, barely above zero. And meanwhile other economies around the world, which performed reasonably well during the dark period from 2008-2010, are finding themselves drawn toward the zero lower bound.

And so a new narrative is gaining adherents (among them Mr Krugman himself): the Hansen-Summers "secular stagnation" story of low rates forever. The story, I write in my column goes like this:

> *[T]he problem is a global glut of savings relative to attractive investment options. This glut of capital has steadily and relentlessly pushed real interest rates around the world towards zero.*
>
> *The savings-investment mismatch has several causes. Dampened expectations for long-run growth, thanks to everything from ageing to reductions in capital spending enabled by new technology, are squeezing investment. At the same time soaring inequality, which concentrates income in the hands of people who tend to save, along with a hunger for safe assets in a world of massive and volatile capital flows, boosts saving. The result is a shortfall in global demand that sucks ever more of the world economy into the zero-rate trap.*

The long downward trend in global real interest rates pre-dates the Great Recession. In the early 2000s the Fed was already struggling to manage a low-rate, low-inflation environment. The glut of global savings in search of safe assets with a reasonable rate of return fueled the American housing bubble. The financial crisis ushered rich-world rates to the zero lower bound, but the fall to zero was probably inevitable. What's more, it is only a matter of time until the rest of the world gets stuck as well:

> *Economies with the biggest piles of savings relative to investment – such as China and the euro area – export their excess capital abroad, and as a consequence run large current-account surpluses. Those surpluses drain demand from healthier economies, as consumers' spending is redirected abroad. Low rates reduce central banks' capacity to offset this drag, and the long-run nature of the problem means that promises to let inflation run wild in the future are less credible than ever.*

This story, which looks increasingly convincing, implies that as the Fed attempts to raise rates the dollar will rise in value and inflation will remain low. The American economy will sputter and stall, forcing a quick reversal in rates – though it might keep growing for a time if the government and households tap the money flowing their way and borrow to fuel consumption (or real estate investment).

If this narrative is the right one, an aggressive-enough central bank can ameliorate the zero-rate problem but cannot solve it. A higher inflation rate would allow an economy to maintain positive nominal interest rates in a world in which the global real rate stays rooted near zero. Modestly ambitious monetary policy that depreciates the currency will boost an economy, but mostly by capturing demand from other countries. Economies that have had enough can adopt capital controls to keep the tide of savings out and regain monetary-policy independence. None of that is especially satisfying. But a proper, sustainable long-run solution would require a fix to the global savings-investment imbalance. That, in turn, might mean dramatic reforms around the world, much higher rates of immigration to rich countries with shrinking workforces, and heavy borrowing by safe-asset issuing governments.

The secular stagnation picture is a grim one. It might be wrong; maybe America's economy will keep on trucking even as the Fed lifts rates. If America finds itself dragged slowly back to zero, the outlook for the global economy over the next decade or so is an uncertain and worrying one.

(*Further reading*) Document VII – Slumponomics Trump and the political economy of liquidity traps, The Economist, Nov 10th 2016

The end of great stagnations

After the Brexit referendum and the election of Donald Trump, and the subsequent market reactions, I think we are closing in on a decent theory of the way liquidity traps end.

That might be going too far. Markets have not had that much time to process the American election outcome, and what time they have had has sent some mixed signals. Yet among the clearest market moves since the morning of November 9th has been a sharp drop in Treasury prices, accompanied by a sharp rise in implied inflation expectations as determined from inflation-protected Treasury securities. Now: sharp is relative. But markets are showing signs of that they expect reflation under Mr Trump. And they have reason to. Mr Trump seems keen on massive tax cuts and a big increase in government spending (on defence, and perhaps also on infrastructure). Mr Trump might just represent macroeconomic regime change.

Let's back up. The theory of liquidity traps first began to develop in the 1930s, when John Hicks picked up where John Maynard Keynes left off, in response to events of the Depression. Keynes explained that when an economy was operating at less than full employment, then a rise in government borrowing could generate a large boost in employment for a small rise in interest rates; rather than crowding out private investment, it would generate more investment by private firms by raising national income. Hicks built on this; when the real interest rate fell to zero and could not be reduced any more, he noted, then expansions in the money supply would cease to be stimulative and fiscal stimulus would be necessary to get the economy moving.

Economic work on the liquidity trap was revived in the late 1990s, when Japan showed that such traps could develop in the modern era. Paul Krugman found, for instance, that monetary policy could be used to escape the trap, provided that the central bank could credibly promise to tolerate high inflation in future. But for central banks to tolerate high inflation at any time is a big ask, and promising it in future is especially hard given the problem of time inconsistency: the high inflation in future would necessarily occur at a time when the economy had managed to escape the trap, at which time the central bank (having gotten the economy out of trouble) would find it tough to resist its natural urge to tighten. What's more, a central bank acting independently to promise high inflation future would risk its independence unless it could be very confident that political leaders were behind it. Which is to say: without clear political support, a big shift in monetary policy is both unlikely to occur in the first place, and unlikely to be seen as credible if, by some chance, it does. Hence, the trap stays the trap.

That leaves fiscal policy. But as Mr Krugman has written in the context of the Great Recession, both fiscal and monetary policy measures might be subject to a "timidity trap". If policy measures, like asset purchases or fiscal stimulus, fail to push an economy all the way out of the liquidity trap, then they may end up discrediting such measures politically, even if they were successful at raising growth relative to a hypothetical no-stimulus case. Most economists reckon that Mr Obama's \$800 billion stimulus raised growth above where it otherwise would have been. It didn't get America out of the low-rate, low-growth slump entirely, however, which allowed Mr Obama's critics to argue, persuasively if not correctly, that the stimulus had not worked at all.

Escaping the liquidity trap means engineering a dramatic rise in expectations for future demand growth. It means a clear departure from past practice: a regime change. And whether the tool used to engineer the shift in expectations is monetary or fiscal, it cannot occur without strong political support.

Now we have a story about the lifespan of a liquidity trap. Once an economy finds itself in one it stays in it until some external force drags it out, or until there is a political change–a break with the past–which enables a macroeconomic regime change. Does this story fit the historical facts? I think so, mostly. In the 1930s, economies got out of their traps when a change in governance led to a abandonment of usual gold- standard monetary-policy practice and/or when the demands of rearmament led to a change in macroeconomic policy. Japan, the first country in the modern era to fall into the liquidity trap, is as close as it has ever been to escaping thanks to the bold economic policies of Shinzo Abe, an elected leader. Brexit, and the choices made by the conservatives under Theresa May, constitute a sharp break with past policy, which has led to a big drop in sterling and a significant rise in expected inflation. Then there is Mr Trump, who might be about to deliver a deficit-financed stimulus. Meanwhile, the other economies stuck in the trap remain stuck.

A key difference between the latter two examples and the earlier ones is that neither Brexit or Trumpism were explicitly about reflating the economy and escaping the slump, in the way that Franklin Roosevelt's decision to leave gold and Mr Abe's Abenomics were. It might therefore be best to characterise Brexit and the American election as political shifts which create the possibility of an escape from the trap, which the respective governments might or might not seize. In America, seizing that opportunity might require the application of pressure on the Federal Reserve. Inflation expectations are not the only thing which has moved since Tuesday; markets reckon the chance of multiple rate hikes in 2017 has also gone up. It does seem important, however, that both of these events are expressions of political radicalism which occurred during slump conditions, and which seem to create the possibility of a departure from those slump conditions. Slump, radicalism, exit seems like something of an historical motif.

Muddling along is a good-enough way to deal with lots of messes, but it is no way to deal with a liquidity trap. Muddling through will continue until the political system can't bear it any more or the outside world forces a change (and history suggests that either endpoint can be very messy). With so many economies now stuck in liquidity traps, it more political shocks seem inevitable. Europe, I would say, will eventually get out of its liquidity trap. Its escape won't be pretty.

(*Further reading*) Document VIII – Why central bankers' conventional tools are no longer working, Lawrence H Summers and Anna Stansbury - The Guardian, New York Times Blog, August 26, 2019

Tweaking inflation targets is not an adequate response to the challenges facing major economies

The world's central bankers and the scholars who follow them are having their annual moment of reflection in Jackson Hole, Wyoming. But the theme of this year's meeting, "challenges for monetary policy", may encourage an insular – and dangerous – complacency.

Simply put, tweaking inflation targets, communications strategies, or even balance sheets is not an adequate response to the challenges now confronting the major economies. Rather, 10 years of below-target inflation throughout the developed world, with 30 more expected by the market, and the utter failure of the Bank of Japan's extensive efforts to raise inflation suggest that what was previously treated as axiomatic is in fact false: central banks cannot always set inflation rates through monetary policy.

Europe and Japan are currently caught in what might be called a monetary black hole – a liquidity trap in which there is minimal scope for expansionary monetary policy. The United States is one recession away from a similar fate, given that, as the figure below illustrates, there will not be nearly sufficient room to cut interest rates when the next downturn comes. And with 10-year rates in the range of 1.5% and forward real rates negative, the scope for quantitative easing and forward guidance to provide incremental stimulus is very limited – even assuming that these tools are effective (which we doubt).

These developments seem to lend further support to the concept of secular stagnation; indeed, the issue is much more profound than is generally appreciated. Relative to what was expected when one of us (Summers) sought to resurrect the concept in 2013, deficits and national debt levels are far higher, nominal and real interest rates are far lower, and yet nominal GDP growth has been far slower. This suggests some set of forces operating to reduce aggregate demand, whose effect has only been partly attenuated by fiscal policies.

Conventional policy discussions are rooted in the (by now old) New Keynesian tradition of viewing macroeconomic problems as a reflection of frictions that slow convergence to a classical market-clearing equilibrium. The idea is that the combination of low inflation, a declining neutral real interest rate, and an effective lower bound on nominal interest rates may preclude the restoration of full employment. According to this view, anything that can be done to reduce real interest rates is constructive, and with sufficient interest-rate flexibility, secular stagnation can be overcome. With the immediate problem being excessive real rates, looking first to central banks and monetary policies for a solution is natural.

We are increasingly skeptical that matters are so straightforward. The near-universal tendency among central bankers has been to interpret the coincidence of very low real interest rates and nonaccelerating inflation as evidence that the neutral real interest rate has declined and to use conventional monetary policy frameworks with an altered neutral real rate.

But more ominous explanations are possible. There are strong reasons to believe that the capacity of lower interest rates to stimulate the economy has been attenuated – or even gone into reverse.

The share of interest-sensitive durable-goods sectors in GDP has decreased. The importance of target saving effects has grown as interest rates have fallen, while the negative effect of reductions in interest rates on disposable income has increased as government debts have risen. Declining interest rates in the current environment undermine financial intermediaries' capital position and hence their lending capacity. As the economic cycle has globalized, the exchange-rate channel has become less important for monetary policy. With real interest rates negative, it is doubtful that the cost of capital is an important constraint on investment.

To take the most ominous case first, with interest-rate reductions having both positive and negative effects on demand, it may be that there is no real interest rate consistent with full resource utilization. Interest-rate reductions beyond a certain point may constrain rather than increase demand. In this case, not only will monetary policy be unable to achieve full employment, it will also be unable to increase inflation. If demand consistently falls short of capacity, the Phillips curve implies that inflation will tend to fall rather than rise.

Even if interest-rate cuts at all points proximately increase demand, there are substantial grounds for concern if this effect is weak. It may be that any short-run demand benefit is offset by the adverse effects of lower rates on subsequent performance. This could happen for macroeconomic or microeconomic reasons. From a macro perspective, low interest rates promote leverage and asset bubbles by reducing borrowing costs and discount factors, and encouraging investors to reach for yield. Almost every account of the 2008 financial crisis assigns at least some role to the consequences of the very low interest rates that prevailed in the early 2000s. More broadly, students of bubbles, from the economic historian Charles Kindleberger onward, always emphasize the role of easy money and overly ample liquidity. From a micro perspective, low rates undermine financial intermediaries' health by reducing their profitability, impede the efficient allocation of capital by enabling even the weakest firms to meet debt-service obligations, and may also inhibit competition by favouring incumbent firms. There is something unhealthy about an economy in which corporations can profitably borrow and invest even if the project in question pays a zero return.

These considerations suggest that reducing interest rates may not be merely insufficient, but actually counterproductive, as a response to secular stagnation.

This formulation of the secular stagnation view is closely related to the economist Thomas Palley's recent critique of "zero lower bound economics": negative interest rates may not remedy Keynesian unemployment. More generally, in moving toward the secular stagnation view, we have come to agree with the point long stressed by writers in the post-Keynesian (or, perhaps more accurately, original Keynesian) tradition: the role of particular frictions and rigidities in underpinning economic fluctuations should be de-emphasized relative to a more fundamental lack of aggregate demand.

If reducing rates will be insufficient or counterproductive, central bankers' ingenuity in loosening monetary policy in an environment of secular stagnation is exactly what is not needed. What is needed are admissions of impotence, in order to spur efforts by governments to promote demand through fiscal policies and other means.

Instead of more old New Keynesian economics, we hope, but do not expect, that this year's gathering in Jackson Hole will bring forth a new Old Keynesian economics.

Lawrence H Summers, US secretary of the treasury (1999-2001) and director of the US National Economic Council (2009-2010), is a former president of Harvard University, where he is currently university professor. Anna Stansbury is a PhD candidate in economics at Harvard University.

(*Further reading*) Document IX – The west must avoid falling into the trap that stifled Japan, Mohamed El-Erian, Teh Guardian, April 9, 2019

US and European policymakers once believed they had the tools to deal with sluggish recovery

Not too long ago, the conventional wisdom held that "Japanification" could never happen in western economies. Leading US economists argued that if the combined threat of weak growth, disinflation and perpetually low interest rates ever materialised, policymakers would have the tools to deal with it. They had no problem lecturing the Japanese about the need for bold measures to pull their country out of a decades- old rut. Japanification was regarded as the avoidable consequence of poor policymaking, not as an inevitability.

And yet the spectre of Japanification now looms over the west. After the 2008 financial crisis the recoveries in Europe and the US were more sluggish and less inclusive than the majority of policymakers, politicians and economists expected. More recently, hopes for achieving "escape velocity" out of the "new normal" of low growth and persistent disinflationary pressure have been dashed in Europe and Japan, and some worry that they may be receding in the US.

Europe, in particular, is back in the grips of a worrying regionwide slowdown. Growth projections have been consistently revised downward and the European Central Bank has acknowledged that its earlier optimism about achieving on-target inflation was misplaced. With yields on government bonds having fallen, the global trade in securities at negative interest rates has reached around $10tn (£7.66tn).

Meanwhile, Japan is approaching its fourth consecutive decade of consistently low nominal growth, inflation and interest rates. In the US, a growing number of economists are worried about a coming slowdown, with some urging the US Federal Reserve to cut interest rates and others calling for it to adopt a higher inflation target in order to combat the risk of excessive disinflation.

Were all those western economists who dismissed the threat of Japanification in the past being too glib? Yes and no. Today's Japanification fears stem from legitimate worries about structural disinflationary forces that could cause lower, less inclusive growth – directly and indirectly. These forces include societal ageing, rising inequality (in terms of income, wealth and opportunity), social and economic insecurity among broad segments of the population, and a loss of trust in institutions and expert opinion.

Along with the zombification of firms after the last asset bubble, these structural factors have led to lower demand, as well as increased risk aversion and self-insurance, rather than growth-promoting risk-pooling, at the margin. Innovation, particularly in artificial intelligence, big data and mobility, is another factor. Though the economic impact of these technologies is ambiguous, there is no doubt that they are reducing entry barriers across a growing number of economic activities and putting downward pressure on prices (the "Amazon effect"), at least in the short term. Nonetheless, their long-term effects on growth and productivity remain to be seen. Growth is also being undercut in less direct ways. For example, persistently low – and in some cases negative – interest rates tend to eat away at the institutional integrity and operational effectiveness of the financial system, thereby reducing bank lending and limiting the range of long-term products that insurance or retirement firms can offer to households. Another indirect effect stems from expectations about the future. The longer growth and inflation remain low, the more tempted households and companies will be to postpone consumption and investment decisions, thus prolonging low growth and inflation.

The western economists who initially underestimated the threat of Japanification did so because they had downplayed or simply ignored these direct and indirect factors. In retrospect, they should not be surprised to find that the societies with the fastest-ageing populations and less inward migration are the ones struggling with Japanification.

Still, those economists were not wrong to argue that policies can play a decisive role in macroeconomic outcomes – especially when structural forces are being amplified by excessive cyclical tightening, as was the case in Japan in 1989. The problem is that they have tended to focus too narrowly on monetary policy, while overestimating its effectiveness. Countries at risk of Japanification need a much broader mix of policies to address the demand side and the supply side of the economy.

Monetary policy, after all, is less effective near the "zero bound" and in scenarios where other "liquidity trap" factors are in play. Large-scale balance sheet operations such as quantitative easing (QE) can buy time by seeking to inject more liquidity directly into the system. But they do not address the underlying issues and they come with their own set of costs, forms of collateral damage and unintended consequences.

The strongest protection against Japanification, then, is a combination of demand- and supply-side measures at the national, regional (in the case of Europe) and global levels. In countries with adequate fiscal space, this could mean looser government budgets and more productivity-enhancing investments (such as in infrastructure, education and training). And in any country facing skills shortages, increased legal migration and better policies to facilitate labour mobility can help close the gap. Moreover, these policies would need to be accompanied by more effective protections for the most vulnerable segments of the population, particularly when it comes to health, training and labour retooling. None of this will materialise without better political leadership and more enlightened global policy interactions.

Japanification offers three lessons that western policymakers and politicians have yet to internalise sufficiently. First, structural pressures make prompt action to reverse low growth, disinflation and zero-to-low interest rates all the more critical. Second, unconventional monetary measures may be necessary but they certainly are not sufficient. And, third, when crafting the required comprehensive policy response, we must recognise that the hurdles are a lot less technical and a lot more political.

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